

The Price of Parenthood: Childcare Costs and Fertility *

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Abstract

Across the developed world, fertility rates have fallen below replacement level, raising concerns over shrinking workforces and ageing¹ populations. U.S. birthrates have reached historic lows, and high childcare costs pose a financial barrier to parenthood. This paper studies how childcare prices shape fertility decisions: whether to have children, when to have them, and how many to have. Using an instrumental variables approach that exploits changes in U.S. state-level childcare regulations that effectively shift the price of childcare, I find that higher prices reduce birth rates, delay first births, and lengthen the interval between first and second births. A 10% increase in the price of childcare leads to a 5.7% decrease in the birth rate (4 births per 1000 women). Declines are largest amongst women aged 30 and above. I propose a theoretical model to explain this age gradient: older women earning higher wages face a greater opportunity cost of their time and thus outsource childcare, making them more sensitive to its price. Consistent with the model's predictions, older parents spend more on formal childcare, and more educated women (as a proxy for income) exhibit greater price sensitivity. Additionally, older mothers are more likely to be considering higher order births, which I find to be more price sensitive.

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¹Please note that I use British English spelling throughout.

1 Introduction

Developed countries across the world are experiencing fertility rate declines. Low fertility rates, absent increased immigration, lead to a shrinking future workforce and an ageing population, which will slow economic growth and strain social services.

Would-be parents cite financial pressures as a key barrier to starting or expanding a family ([Institute for Family Studies, 2022](#)), with childcare one of the earliest and largest costs of child-rearing. Childcare costs represent a substantial financial burden in the U.S; families spend 9% to 16% of median income on full-time care for one child alone ([U.S. Department of Labor, 2024](#)), and low-income families are particularly strained. Furthermore, high childcare costs can push mothers to reduce their working hours or drop out of the labour market ([Haan and Wrohlich, 2011](#)).

In this paper, I study how childcare prices affect parents’ decisions to have children, how many to have, and when to have them. To my knowledge, this paper provides some of the first quasi-experimental evidence on the causal relationship between childcare prices and fertility in the U.S. setting. Prior work on the cost of childcare is limited, particularly in the U.S., reflecting the difficulty of finding plausibly exogenous cost variation.

To answer this question, I assemble a dataset on U.S. childcare regulations to construct an instrument for childcare prices. I compile state-level data on maximum group size and minimum staff-child ratios for 2010 to 2022, collecting data directly from historical state administrative codes and licensing regulations to address gaps and errors in existing sources. These regulations affect prices by limiting capacity and dictating staffing requirements. I link these data to administrative birth records and the National Database of Childcare Prices (NDCP). My identification strategy exploits regulatory changes in Delaware, Nevada, and Vermont over this period, generating plausibly exogenous variation in childcare prices.

I document key facts about childcare prices in the U.S., which are substantial for many families. The average cost of childcare as a share of median household income ranged from 5% to 35% across counties in 2022. There is significant geographical variation in childcare prices, with higher prices on the West Coast and in the Northeast. However, plotting the annual cost of childcare as a share of median household income shows that there are areas throughout the U.S. where childcare costs represent more than 20% of median household income.

I show that childcare facility regulations significantly impact the price of childcare for children under three. This relationship is a useful input into policy debates around regulating the childcare market. It also indicates that these regulations are relevant instruments for prices. A one-child increase in the maximum group size (6% of the mean) reduces the weekly price of childcare by \$1. This effect is larger in lower-income counties, where parents are more likely to demand cheaper childcare, so providers are more likely to be operating at the binding maximum group size. In below-median income counties, a one-child increase reduces weekly childcare prices for children under three by \$5.40, roughly five times the price reduction estimated in above-median income counties.

The core of this paper analyses the effects of childcare prices on birthrates. A simple Ordinary Least Squares (OLS) regression of birthrates on childcare prices would likely suffer from statistical endogeneity. Price changes can reflect either a supply or demand response, or both. Childcare prices changes can respond to shifts in births and can impact the choice to conceive. Such simultaneity would bias my estimates towards zero, and indeed, I find null effects with OLS. To overcome endogeneity, I instrument childcare prices with the childcare facility regulations.

My quasi-experimental results show that fertility is sensitive to childcare prices. I find that a 10% increase in the price of childcare for 0 to 2-year-olds leads to a 5.7% decrease in the birthrate of women aged 20 to 44 (4 births per 1000 women). Births to White, Black, and Hispanic mothers are all price sensitive. An increase in the maximum group size by 10 children leads to a rise in birthrates of 3.7%. I also find that a rise in childcare prices leads mothers to delay their first birth, and increase the time between their first and second birth. A 10% increase in the childcare price shifts age at first birth by 4 months and the time between the first and second birth by half a month.

These estimates are robust to a battery of sensitivity analyses and alternative approaches. I defend the validity of my instrument by assessing the plausibility of the identifying assumptions. One potential concern with this identification strategy is that the childcare regulations may affect the quality of childcare, and quality may directly impact fertility decisions. To respond to this concern, I first show that the results are robust to controlling for a measure of quality: staff turnover. I also conduct a bounding exercise. If quality does in fact impact birthrates, I show that my Instrumental Variables (IV) estimates represent a lower bound. I decompose the size and components of the bias, and estimate the size of the true population parameter using a another measure of childcare quality: the share of childcare providers that

have received accreditation.

I show that birthrate responses to price changes vary by age. Older women, aged 30 and above, are more price responsive than younger women. I outline a simple theoretical model that provides a framework for understanding this age gradient. One may expect younger women to be more price responsive than older women, given that they are more likely to be lower-income, with childcare representing a larger share of their budget. However, I show that older mothers can be more price sensitive than younger mothers. Through human capital accumulation from work experience, older women earn a higher wage, and so the opportunity cost of their time is higher. Women earning higher wages will outsource a larger share of childcare, if not all. High levels of outsourcing make mothers more exposed to increases in the price of childcare, driving increased price sensitivity. Younger, lower-income mothers, on the other hand, are more likely to reduce their working hours and provide more childcare themselves. Additionally, second births to older mothers are more price sensitive simply because the total spending on childcare rises.

I test my model predictions and demonstrate several potential mechanisms driving the age heterogeneity. First, using Consumer Expenditure Survey data, I confirm that older parents do spend more on formal childcare. Second, higher-earning women in this setting are more price sensitive. Using education as a proxy for income, I show that birthrates for women with an undergraduate degree are more price sensitive than those for women without. Finally, I provide suggestive evidence that second and third birthrates are more price sensitive than first birthrates.

This paper builds on several strands of the literature. Broadly, by studying childcare prices, I contribute to the literature studying how financial barriers and incentives affect fertility. Child subsidies ([Cohen et al., 2013](#); [Milligan, 2005](#)), cash transfers ([González, 2013](#); [Ang, 2015](#)), paid maternity leave ([Raute, 2019](#)), housing credits ([van Doornik et al., 2024](#)), and tax incentives ([Laroque and Salanié, 2014](#); [Whittington et al., 1990](#)) have pro-natal effects. Car seat laws that require parents with three children to purchase a larger car also reduce fertility ([Nickerson and Solomon, 2024](#)). Welfare reforms have null to small effects on childbearing ([Kearney, 2004](#); [Rosenzweig, 1999](#); [Joyce et al., 2003](#); [Moffitt, 1998](#)), except for a reform that reduced welfare for immigrants ([Amuedo-Dorantes et al., 2016](#)). Economic shocks ([Kearney and Wilson, 2018](#)), policy interest rates ([Cumming and Dettling, 2024](#)), and families' assets ([Dettling and Kearney, 2014](#)) have also been found to causally impact birthrates.

The causal effects of childcare costs on fertility remain understudied outside of the structural

modelling literature.² In European settings, where public provision of childcare is more common and spending on policies to support families is high, prior research has documented notable fertility responses to childcare policy reforms. The introduction of universal childcare in Germany led to an expansion in fertility, for second and third births in particular (in line with my findings) (Bauernschuster et al., 2016). A reform that capped childcare costs increased first births in Sweden (Mörk et al., 2013). Using simulation models, Rindfuss et al. (2010) estimate that increased availability of childcare spaces in Norway would lead to higher childbearing. These papers focus on policy reforms that shift the cost of childcare. I instead use an IV approach and granular price data to estimate the effect of childcare prices directly, which allows me to quantify price elasticities.

While important and informative, the conclusions and magnitudes from these papers may not carry across to the U.S., where childcare is chiefly delivered through the private market, the responsibility for paying for childcare is placed on parents (Davis and Sojourner, 2021), and the population is more racially diverse. The U.S. spends 0.33% of GDP on public early care and education for 0 to 5-year-olds relative to 1.3% or more in France, Norway and Sweden and the OECD average of 0.74% (OECD, 2019).³ The cost of childcare as a share of household income is substantially higher in the U.S. than other OECD countries (OECD, 2022). Early work by Blau and Robins (1989) suggests that childcare costs are birthrate reducing in the U.S. Averett and Wang (2023) find that the Child and Dependent Care Tax Credit (CDCTC) has no effect on fertility, which the authors attribute to an increase in female labour force participation. Evidence from the marketisation of childcare in the U.S., though, suggests that the relationship between childcare costs and fertility rates holds. The marketisation of childcare and immigrant inflows has allowed educated women to remain in the labour force and work longer hours (Cortes and Tessada, 2011). A few authors explore the fertility responses (Furtado, 2016; Furtado and Hock, 2010; Bar et al., 2018; Hazan and Zoabi, 2015), finding that immigrant inflows are associated with an increased likelihood of having a child and that changes in the relative cost of childcare can explain highly educated women’s ability to have more children and increase their working hours. To this empirical literature, I contribute new causal evidence on the childcare price elasticity of fertility in a

²There is a large structural literature that models female labour force participation and fertility decision making (e.g. Keane and Wolpin (2010); Blundell et al. (2016)), some of which also study childcare costs or policies (e.g. Haan and Wrohlich (2011); Bick (2016); Guner et al. (2024)). These papers find that reducing the cost of childcare encourages fertility, except for when subsidies require tax increases.

³Public expenditure on early-childhood education and care defined as all public spending towards formal day-care services and pre-primary education services. Data adjusted for cross-national differences in school starting age.

setting where access to high-quality, affordable childcare is limited for most.

The decline in U.S. fertility rates is driven by both an increase in childlessness and declines in total number of children, and observed across demographic groups. Replicating the fertility rates of cohorts before 2007 would necessitate more births occurring after age 30 among today’s women ([Kearney and Levine, 2021](#)). My results suggest that birthrates, and higher order birthrates in particular, are sensitive to childcare price changes. These findings indicate that containing childcare costs is important for policymakers seeking to mitigate continued fertility declines.

2 Background

In the U.S., there is limited public involvement in the provision of childcare. The childcare market is mostly made up of small private businesses (both profit and non-profit) ([Tekin, 2021](#)) and can be divided into formal and informal markets.

Formal childcare settings include day care centres, preschool, nurseries and family childcare homes. Centre-based care options are usually provided by businesses, places of worship, or community-based organisations. Centre-based childcare is typically split up into classrooms by child age (infants, toddlers, preschool, school-age) with teaching staff members overseeing the children ([Brown and Herbst, 2022](#)). High-quality, centre-based care has been found to have positive effects on children’s education, future earnings ([Bailey et al., 2021](#); [Garces et al., 2002](#)) and health ([Carneiro and Ginja, 2014](#)). However, there is a low supply of high-quality care and it is often the most expensive option for parents. In family childcare homes, the childcare takes place in the provider’s home and children are typically in mixed age groups. Often, the provider also cares for their own children alongside the other children.

Formal childcare facilities are licensed and regulated by state and federal governments to ensure that they meet minimum health and safety requirements. These requirements span a range of domains, including building safety, sanitation, health, background checks, and staff qualifications, training, and supervision. They are designed to support child safety, well-being and development. Family childcare homes tend to face less stringent regulations than childcare centres.

Informal childcare is care that is unlicensed and unregulated, and provided in the child or

caregiver’s home by nannies, babysitters, au-pairs, family members or friends. Informal care can often be cheaper, particularly if there are multiple children being cared for at the same time, and a more flexible option for mothers and families.

Public funding for childcare is targeted at low-income families and the majority of funds come from the federal government. The main childcare assistance programme is the Child Care and Development Fund (CCDF), a subsidy scheme for low-income households that is both federal- and state-funded. Eligibility is based on income and child age (children must be under 13). Furthermore, households must need the childcare to work or engage in work-related activities ([Tekin, 2021](#)). In 2022, 1.4 million children under age 13 were served by the CCDF ([Administration for Children and Families, Office of Child Care, 2022a](#)), with 28% of these children under age three ([Administration for Children and Families, Office of Child Care, 2019](#)). This implies that 4% of children under three receive some level of CCDF subsidy. 96% of children served were funded through vouchers ([Administration for Children and Families, Office of Child Care, 2022c](#)), which parents can use to purchase private childcare from their choice of provider (formal and informal). Of income-eligible families served by the CCDF, 31% paid no additional copayment for their childcare ([Administration for Children and Families, Office of Child Care, 2022b](#)).

Government funding for childcare is also directed towards the Temporary Assistance to Needy (TANF) and Head Start programmes for low-income families. Head Start is the only federal public provider of childcare, and serves a small proportion of the population. Head Start associated programmes (Head Start, American Indian and Alaska Native Head Start, and Migrant and Seasonal Head Start) served 249,094 children under three in 2023 ([Administration for Children and Families, 2023](#)), or 2% of the U.S. under three population ([U.S. Census Bureau, 2025b](#)).

Parents who pay for childcare to work or search for work are eligible to claim a tax credit of up to \$1,050 for one child, or \$2,100 for two or more children (under 13).⁴ Parents can also save money for childcare services in a Dependent Care Flexible Spending Account (DCFSA) – if offered by their employer. A DCFSA is a pre-tax benefit account that can be used for spending on before and after school care, babysitting, nannies, formal childcare, and summer day camps. Both parents must be working, looking for work, or full-time students.

⁴The percent of childcare expenses that is eligible for the tax credit varies by household income and subject to a maximum that depends on the number of children. Households with gross income of up to \$15,000 can claim 35% of \$3,000 (one child), or \$6,000 (two or more children) Households with gross income of \$43,001 and over can claim 20% of these same maximum amounts.

Data from the Early Childhood Program Participation (ECP) component of the National Household Education Surveys (NHES) in 2019 show that 53% of children under 3 participate in some form of non-parental care or programme arrangement, for an average (mean) of 16 hours a week. Amongst these children, for their primary form of care, 37% went to a centre-based programme, 13% received care from a relative, 40% were in a family childcare home, and 6% received other non-relative care (e.g. friends or nannies).

3 Data and sample construction

3.1 Data

3.1.1 Birthrates

My source of data on fertility is the restricted use National Vital Statistics System (NVSS) birth records data. The NVSS contains information on the universe of births in the U.S., making it the most comprehensive source of fertility data available. These administrative data are derived from birth certificates and contain detailed demographic information on the mother (e.g. marital status, age, race, ethnicity, and birth history). I use these data to conduct individual and county-level analysis. The NVSS has county identifiers, so I combine these data with Census population counts to calculate age-specific county-level birth rates for the years 2010 to 2022 inclusive. For the heterogeneity analysis by education, I calculate population counts for women with and without an undergraduate degree by age band using the 5-year American Community Survey (ACS). Fertility decisions take place at the time of conception, rather than at birth. I calculate the estimated date of conception by subtracting the mother's gestation length in weeks from the midpoint of the month of birth. Using the individual-level data I can also explore whether potential mothers delay births or increase time between births (in order to save money) in response to higher costs of childcare.⁵

I exclude 2020 due to the global COVID-19 pandemic. Childcare facilities were closed or had reduced capacity during the pandemic, and stay-at-home orders and the rise of remote work influenced fertility decisions ([Adelman et al., 2023](#)).

⁵Note that I cannot link mothers across births.

3.1.2 Childcare prices

The childcare price data is from the National Database of Childcare Prices (NDCP), which provides county-level prices for formal childcare by child age. I use the median price of full-time centre-based care, aggregated for 0 to 2-year-olds inclusive. I focus on prices for children under 3 as these are the initial costs that prospective parents would face.

The NDCP is collated by the U.S. Department of Labor (DOL) Women’s Bureau (WB) and sourced from historic market surveys conducted by U.S. states. States have collected data on childcare market prices since 1998, as a requirement by the Administration for Children and Families (ACF) at the U.S. Department of Health and Human Services. The ACF needs these data to calculate state reimbursement rates for the CCDF grants, and in order for states to receive funding they must provide information on the prices charged at childcare facilities within their state. Thus, it is in the states’ interest to collect these data. The ACF demands that states conduct the surveys every three years (this figure was two years until 2016), but some states provide annual data. The market surveys focus on the regulated, formal childcare market, and data is most complete for childcare centres. Despite the requirement to collect these data, and the incentives to do so, there are missing data at the county, year, and child-age level. Furthermore, states varied in their approach to collecting these data, which has resulted in data missing-ness and aggregation. To address the missing-ness, the DOL WB imputed data for missing years, counties, and ages. I discuss how I handle the imputed data in subsection 3.2. Some states appear to aggregate data across counties at a childcare “market” level. I discuss the implications of this aggregation in Section 5.3.

3.1.3 Childcare regulations

My empirical strategy relies on state-level regulations that shift the cost of providing childcare. A public dataset that tracks the regulations placed on formal childcare facilities by year across my study period does not exist; the Child Care Licensing Studies data used in [Araujo et al. \(2025\)](#) and [Lee et al. \(2024\)](#) are not collected annually, and only record whether a regulation has changed since the prior survey wave, which is not well suited for my empirical strategy. I assemble a hand-collected dataset on childcare facility regulations. Specifically, I collect the maximum group size (the maximum number of children allowed in a room) and minimum ratio (the ratio of children per staff member) by age group of child for each U.S. state-year in my sample. This data collection process requires finding historical copies of

the administrative code or childcare facilities licensing regulation for a state. In addition, I use legal research resources such as “Casetext” to help identify the history of law changes. With the regulations in hand, I then search the text for any references to the a) maximum group size, and b) ratio for childcare centres, and extract the data for these two regulations by the age group of the child.

3.1.4 Additional datasets

I supplement these data with the 5-year American Community Survey (ACS) to capture information on county-level socioeconomic and demographic characteristics, and the U.S. Federal Housing Finance Agency House Price Index to control for single-family house prices.

For exploring mechanisms, I rely on data from the U.S. Bureau of Labor Statistics Consumer Expenditure Surveys (CES). The CES provides detailed information on categories of expenditures for individual survey respondents, as well as demographic characteristics of these respondents. I use the 2010 and 2022 surveys, as these years bookend my study period. In these data I can observe spending on childcare centres by respondent age. I define spending on childcare centres as any spending tagged with the uniform commercial code of “670310” for day care centres, nursery, and preschools.

For evaluating robustness and analysis of the impacts of the childcare facility regulations on childcare markets, I utilise data from the Census County Business Patterns (CBP) and Quarterly Workforce Indicators (QWI) series. The CBP provides data on the number of establishments at various geographical levels, disaggregated by industry. The CBP is derived from the Census Business Register, which tracks all establishments with paid employees in the U.S. The QWI contains data on local labour market conditions by industry. It is sourced from the Longitudinal Employer-Household Dynamics linked employer-employee micro-data. I restrict these datasets to the childcare industry, conditioning on NAICS codes “6244//” for ‘Child Day Care Services’. I aggregate the QWI quarterly data to the annual level by taking the mean, as advised in the QWI methodology guide.

For robustness analysis, I use the CBP data and Census population estimates to calculate the share of childcare establishments per 0 to 5 year old. Additionally, I estimate the share of high-quality childcare establishments, which I describe in more detail in [Appendix A](#). I use the QWI data to obtain a measure of childcare quality: staff turnover.

3.2 Sample construction

The birth records sample consists of women aged 20 to 44 who gave birth between 2010 and 2019, 2021 or 2022 inclusive. I then create a county-level sample for analysing birth rates, and individual-level samples to analyse birth timing decisions.

In order to accurately measure when birthing decisions are made and the prices facing potential mothers, I use the non-imputed childcare price data. I restrict my sample to state-years for which the NDCP data is not imputed on the following criteria: year, geography, and child age. First, I exclude imputed years of data from my sample as they will not reflect the actual prices facing mothers in those years. This is particularly important given that my empirical approach relies on childcare regulatory changes over time. Second, I drop observations where the county-level prices are imputed from state-level data, to ensure that I am capturing the market-area prices facing potential mothers. A handful of states only provided the price data at the state level. However, there is sufficient geographical variation in prices within the other states to conduct more granular analysis. Third, I exclude observations that impute data based on child age. To impute missing prices for age groups, the DOL WB assigned the price for a different age group. As I focus on childcare prior to age 3, I don't want imputed prices that reflect older ages (particularly school age) to be in my analytical sample.

This exercise leaves me with data on births and birthrates for 30 states,⁶ 1,415 unique counties, and 7,495 county-years for the main county-level analysis. For the individual-level analysis, I create separate samples of first, second and third time mothers to analyse birth spacing decisions for these subgroups. I drop any second (third) births that occur less than 9 months after the first (second) birth. I also remove multiple births to ensure that I am comparing births of the same parity in my first, second, and third birth samples. I have 4,178,730 first time mothers, 3,418,884 second time mothers, and 1,864,605 third time mothers.

Table 1 shows summary statistics for my sample of counties. The sample counties have lower earnings and more unemployment than the whole U.S. The weighted mean earnings in the sample is \$32,969, and the median is \$28,266. In 2022, the median sample earnings were \$38,125. The median for the whole U.S. in 2022 was \$47,960 ([U.S. Census Bureau, 2023](#)).

⁶Alaska, Alabama, Arizona, California, Connecticut, Delaware, Florida, Idaho, Illinois, Kansas, Kentucky, Louisiana, Massachusetts, Maryland, Maine, Minnesota, Nevada, Ohio, Oregon, Pennsylvania, South Carolina, South Dakota, Tennessee, Texas, Utah, Virginia, Vermont, Washington, and Wisconsin.

The unemployment rate for the whole U.S. in 2022 was 3.6 percent of the labour force ([U.S. Bureau of Labor Statistics, 2023](#)). In the sample counties, the weighted mean is 6.9 percent.

The sample is broadly representative of the wider U.S. population with regards to race and ethnicity. Across the whole U.S., in 2022, 75% of the population identified as White, 14% identified as Black, 6% identified as Asian, and 19% identified as Hispanic ([U.S. Census Bureau, 2025a](#)). The sample weighted mean share of Whites is 73%, for Black individuals the share is 11%, for Asians the share is 5% and for Hispanics, 21%.

The sample mean childcare price for full-time care at a childcare centre for 0 to 2-year-olds in Table 1 is winsorised at the 99th percentile, as this is what is used in later analysis, and adjusted for inflation using base year 2010. The unadjusted mean childcare price in 2022 was \$200, and the median \$187. For 2010, these figures were \$143 for the mean and \$137 for the median.

Table 1. Descriptive statistics

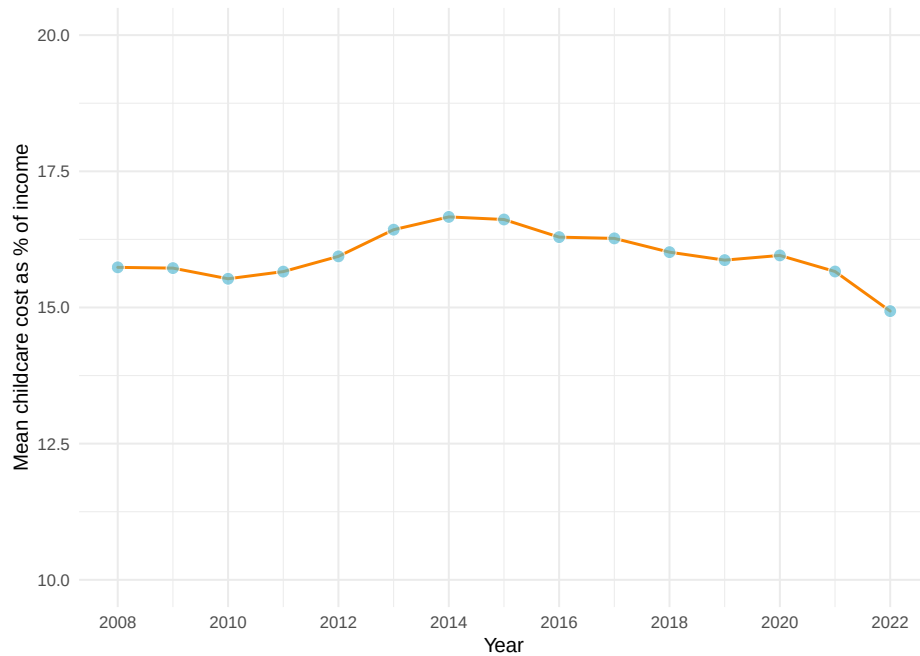
	Mean	Weighted Mean	Median	St. Dev.	Min	Max
Total Births per 1000 (20-44)	69.76	65.49	71.04	18.02	1.93	177.30
Maximum Group Size	16.10	17.87	13.08	7.55	8.00	30.00
Staff-to-Child Ratio	0.20	0.20	0.20	0.03	0.14	0.33
Childcare Price	146.57	189.07	137.75	48.89	69.71	310.24
Earnings	29,544	32,969	28,266	6,414	4,725	69,937
Median Household Income	52,041	60,396	49,303	14,000	22,333	136,268
Unemployment Rate	6.57	6.88	6.20	2.83	0.10	22.70
Male-Female Ratio	1.07	1.02	1.03	0.17	0.76	3.12
Male LFP (%)	79.68	83.18	81.60	9.19	16.00	96.90
Female LFP (%)	70.74	72.44	71.00	6.89	39.90	94.10
White (%)	85.18	73.43	89.40	12.74	15.00	99.50
Black (%)	7.20	11.30	2.70	10.82	0.00	82.20
Asian (%)	1.53	5.15	0.70	2.60	0.00	35.90
Hispanic (%)	11.18	21.29	4.70	15.41	0.00	95.60

Notes: N = 7,495. Data: NVSS birth records, SEER county population counts, NDCP childcare prices, hand-collected maximum group size data, American Community Survey 5-year estimates, for 2010-2019, 2021-2022. “Total Births per 1000 (20-44)” is the county-level birthrate per 1000 women aged 20 to 44. “Maximum Group Size” is the maximum group size average for 0-2 year olds. “Staff-to-Child Ratio” is the average staff-to-child ratio for 0-2 year olds. “Childcare Price” is the median weekly price for full-time care at a childcare centre averaged for 0 to 2 year olds, adjusted for inflation using base year 2010 and winsorised at the 99th percentile. “Earnings” is the median earnings for the population aged 16 and above. “Median Household Income” is the median household income. “Unemployment rate” is the unemployment rate of the population aged 16 and above. “Male-Female ratio” is the ratio of men to women aged 20 to 49 years old. “Female LFP (%)” is the labour force participation rate of the female population aged 20 to 64 years old. “Male LFP (%)” is the labour force participation rate of the male population aged 20 to 64 years old. “White (%)” is the percent of the population that identifies as White. “Black (%)” is the percent of the population that identifies as Black. “Asian (%)” is the percent of the population that identifies as Asian. “Hispanic (%)” is the percent of the population that identifies as Hispanic or Latino regardless of race. Weighted means weighted by population.

3.3 Descriptive patterns in the cost of childcare

I document several patterns in the cost of childcare in the U.S. between 2010 and 2022. First, although the price of childcare has risen over time, the estimated annual cost of childcare for 0 to 2-year-olds as a share of median household income has not changed markedly since 2010 (see Figure 1).

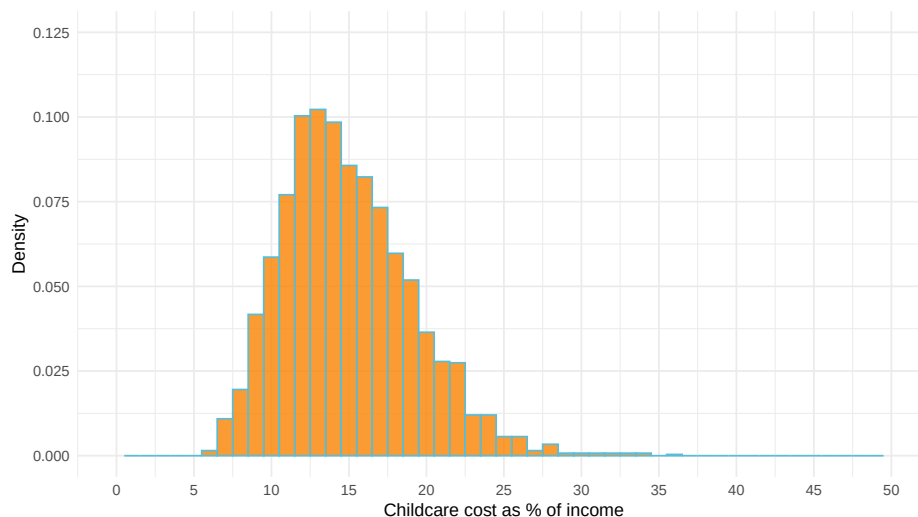
Figure 1. Avg. annual cost of full-time centre-based childcare for 0-2 years as a share of median household income over time



Notes: Data: National Database of Childcare Prices (including imputed data), American Community Survey, for 2010-2022. This plot shows a time series of the average annual cost of full-time care at a childcare centre for 0 to 2-year-olds, as a percentage of median household income at the U.S. county level.

In most counties, the annual cost of childcare falls between 10 to 20% of median household income, but can reach up to 35% of median household income. Figure 2 plots the distribution of the annual cost of childcare as a share of household income for 2022 (See Appendix Figure E1 for the 2010 plot).

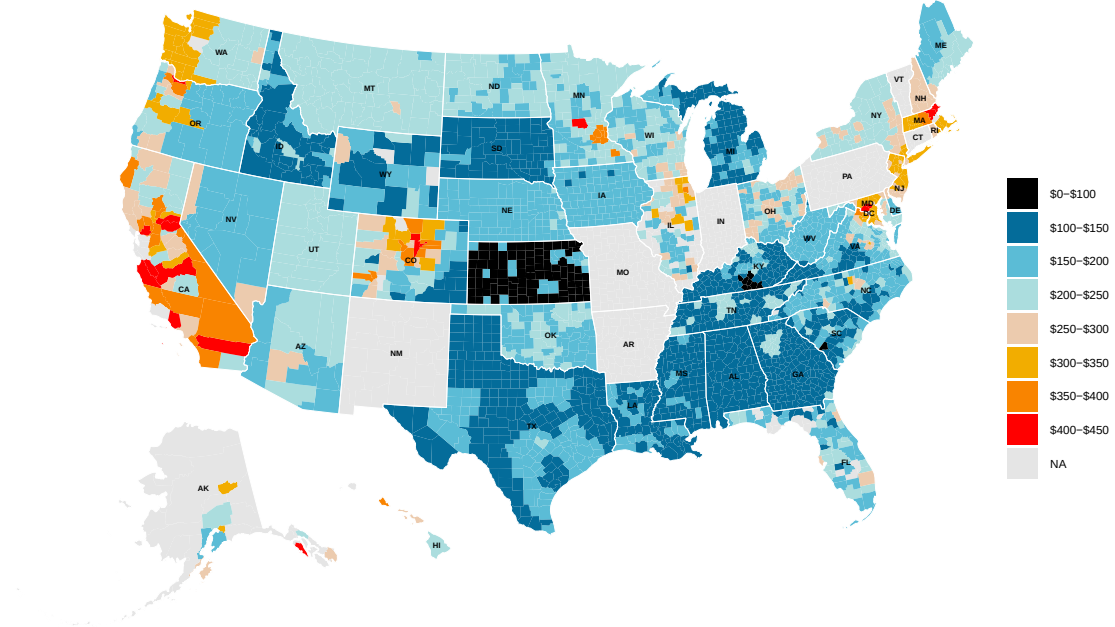
Figure 2. Distribution of avg. annual cost of full-time centre-based childcare for 0-2 years as a share of median household income, 2022



Notes: Data: National Database of Childcare Prices (including imputed data), American Community Survey, for 2022. The plot shows the distribution of the average annual cost for full-time care at a childcare centre for 0 to 2-year-olds, as a percentage of median household income at the U.S. county level.

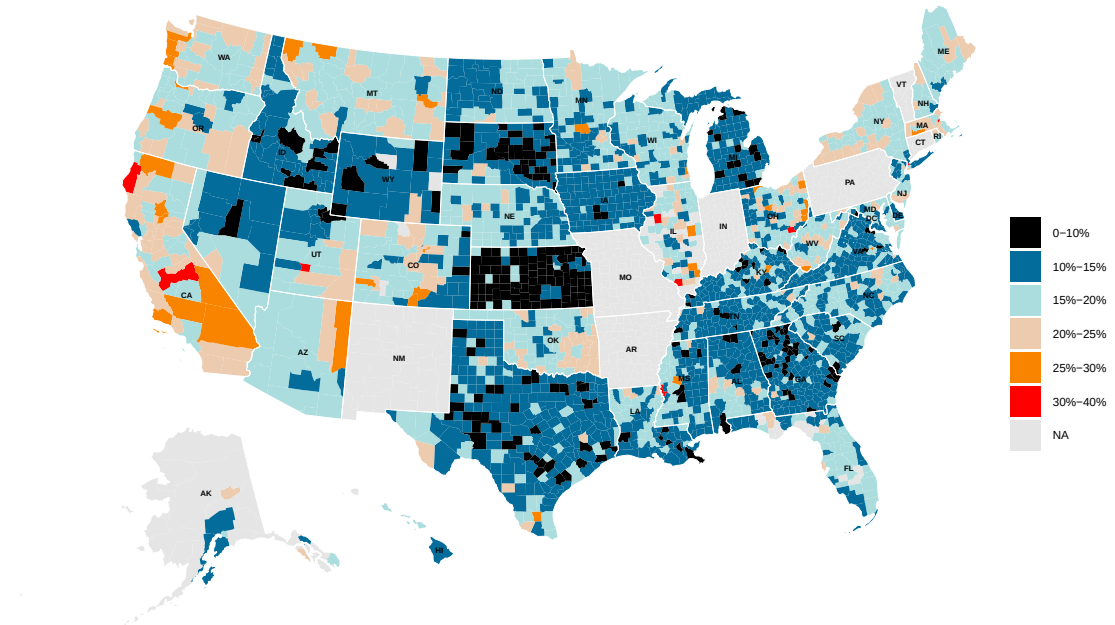
There is substantial geographical variation in the cost of childcare, with parents living on the West Coast and in the Northeast facing much higher prices. Figure 3 shows the geographical distribution of childcare prices for full-time centre-based care for children under 3 across the U.S. in 2022. We also observe this pattern in the geographical distribution of the annual cost of childcare as a share of median household income, but to a lesser extent. There are counties spread across the U.S. where this share is at 20% or higher. Figure 4 shows a map of the annual cost of childcare as a percentage of median household income.

Figure 3. Avg. weekly price of full-time centre-based childcare for 0-2 years, 2022



Notes: Data: National Database of Childcare Prices (including imputed data), for 2022. This map shows the median weekly price for full-time care at a childcare centre averaged for 0 to 2-year-olds, by U.S. county.

Figure 4. Avg. annual cost of full-time centre-based childcare for 0-2 years as a share of median household income, 2022



Notes: Data: National Database of Childcare Prices (including imputed data), American Community Survey, for 2022. This map shows the average annual cost for full-time care at a childcare centre for 0 to 2-year-olds, as a percentage of median household income, by U.S. county.

4 Empirical strategy

I evaluate the causal effect of childcare prices on birth outcomes using a two-stage least squares (2SLS) approach. A naive OLS regression of birth outcomes on childcare prices could suffer from statistical endogeneity. One source of such endogeneity is simultaneity. Childcare prices are determined in equilibrium, thus price changes can reflect either a supply or demand response (or both). A lower birth rate would lead to reduced demand for childcare and could lead to lower prices, and/or a shock to the cost of childcare could increase prices and reduce the birth rate. Measurement error may also arise from imprecise measurement of childcare prices. As discussed in Section 3, the NDCP dataset is constructed from state surveys of childcare facilities. We can imagine that this is an imperfect process, with limitations on collecting fully accurate information. Omitted variables bias may also be a concern with OLS. Career-focused parents with low fertility intentions may sort into high-cost, high-amenity urban areas, where childcare prices are likely also high. To overcome these issues, I instrument childcare prices with regulations placed on formal childcare facilities.

Regulations placed on childcare facilities affect the cost of providing childcare. Increasing health and safety requirements, setting stricter staff-child ratios and limiting the number of children allowed in a room can all raise the cost of running a childcare centre. In the main analysis, I instrument the price of childcare in a county with the state-mandated maximum number of children permitted in a room, the maximum group size, in a given year. To test the robustness of the results, I use both the state-mandated maximum group size and staff-child ratio as instruments. These regulations apply to all formal childcare centres and I use the average group size and staff-child ratio for children less than 3 years old. When the mandated maximum group size is reduced, the childcare facility has to decrease the number of children they can cater for, which can lead the childcare price to rise. When the staff-child ratio increases, fewer children per staff member are permitted, which can also lead the price to increase. This is the intuition behind the instruments.

My IV identification strategy rests on two assumptions: relevance and the exclusion restriction. I establish relevance and address the exclusion restriction in Section 5.

4.1 Specification

The empirical model for analysing effects on birth rates is described by the following equations:

$$\ln(\text{Birthrate})_{ct} = \zeta + \beta \cdot \ln(\text{PriceofChildcare})_{ct} + \mathbf{X}_{ct} \cdot \eta + \gamma_{1t} + \delta_{1c} + \epsilon_{ct} \quad (1)$$

$$\text{PriceofChildcare}_{ct} = \alpha + \mu \cdot Z_{st} + \mathbf{X}_{ct} \cdot \kappa + \gamma_{2t} + \delta_{2c} + \varepsilon_{ct} \quad (2)$$

The subscript c denotes county and t denotes year. Z_{st} is the state-level maximum group size in place for childcare centres, or both the maximum group size and the staff-child ratio, averaged for 0 to 2-year-olds. I control for time varying county-level characteristics X_{ct} , which includes median earnings, the unemployment rate, female labour force participation, the male-female ratio, the house price index, and the racial and ethnic composition (Black, White, Asian and Hispanic) of the county. I control for county fixed effects δ_c and year fixed effects γ_t to account for time-invariant differences across counties and time trends. I cluster standard errors at the state-year level.⁷ I weight estimates by the number of births, in line with the literature. The coefficient of interest is β , which captures the local average treatment effect (LATE). The LATE is the causal effect of the cost of childcare on birthrates in counties where the regulations led to a change in prices for formal childcare, weighted by the change in prices.

The empirical model for analysing effects on individual mother i 's birth timing outcomes is given by:

$$Y_{ict} = \zeta + \beta \cdot \ln(\text{PriceofChildcare})_{ct} + \mathbf{X}_{ict} \cdot \eta + \gamma_{1t} + \delta_{1c} + \epsilon_{ict} \quad (3)$$

$$\text{PriceofChildcare}_{ct} = \alpha + \mu \cdot Z_{st} + \mathbf{X}_{ict} \cdot \kappa + \gamma_{2t} + \delta_{2c} + \varepsilon_{ict} \quad (4)$$

Y_{ict} is either age at first birth (in years), time between the first and second birth (in months), or time between the second and third birth (in months). Z_{st} is as before. I control for the same county-level characteristics as in Equations 1 and 2.⁸ In addition, I control for the following additional individual-level characteristics: race and ethnicity (non-Hispanic White, non-Hispanic Black, Hispanic), marital status, and month of birth. Again, I include county (δ_c) and year (γ_t) fixed effects and cluster standard errors at the state-year level.

⁷Appendix Table D1 shows that the results are robust to clustering at the state level, but doing so makes the number of clusters small given that I have 30 out of the 50 states in my sample.

⁸In the individual-level analysis I do not control for racial and ethnic composition at the county level. Instead I control for these variables at the individual level.

The coefficient of interest β captures the local average treatment effect. Here, the LATE is the causal effect of the cost of childcare on birth timing outcomes amongst mothers living in counties where the regulations led to a change in prices for formal childcare, weighted by the change in prices.

This approach exploits variation in the presence and strength of childcare facility regulations across time and states. In my data, I observe regulatory changes in three states: Delaware (2011), Nevada (2017), and Vermont (2016). For the main analysis with the maximum group size as the instrument, I exploit regulatory changes in Nevada and Vermont. In 2016, Vermont relaxed group size requirements for 18 month olds. In 2017, Nevada introduced group size requirements for the first time and increased staff-child ratios for 18 month and 3-year-olds. For the robustness analysis with both instruments, I exploit changes in Nevada, Vermont, and Delaware. In 2011, Delaware introduced group size requirements for the first time. Although the two-instrument specification delivers a stronger first stage and similar point estimates, I adopt the single-instrument specification as my main approach, as it proves more stable across a range of robustness checks. Across my study period there were four more states (Arizona, Virginia, Louisiana, Utah) that made changes to the mandated maximum group size or staff-child ratio for children under three, but due to missing data in the NDCP (as described Section 3) I do not observe price data before and after the regulatory changes in these states.

Regulation is more likely to bind in counties where childcare providers are not already exceeding the regulatory threshold. Providers who already operate above the mandated group size or staff-child ratio, to differentiate their offering, charge higher prices, or meet benchmarks set by associations such as the NAEYC, face no cost from a change in the threshold. Changes in prices following a change in regulation reflect a weighted average of responses across providers. Yet even when only a subset of providers is directly constrained, market prices can still adjust in equilibrium, as constrained providers change capacity or raise prices, with ensuing responses from the rest of the childcare market. One hypothesis I test in Section 5.2.1 is whether the strength of the price response differs by county income, as providers in low-income counties are more likely to operate close to the regulatory minimum in order to keep childcare affordable.

5 Results

5.1 Effect of regulations on childcare prices

Table 2 shows that an increase in the maximum group size reduces childcare prices; an increase by one child reduces the weekly childcare price for 0 to 2-year-olds by \$1.05. I show both the raw childcare price (adjusted for inflation using base year 2010) and the log of the childcare price as outcome variables. The former is simpler to interpret as a first stage, but the latter is what is used in the IV analysis. Both specifications deliver a large F statistic, providing evidence of a strong first stage and instrument relevance.

Table 2. First stage

	Childcare Price	
	Price	Log(Price) $\times 100$
Maximum Group Size	-1.06*** (0.21)	-0.65*** (0.13)
Mean (Childcare Price)	146.57	146.57
Mean (Maximum Group Size)	16.10	16.10
F-stat	24.59	25.76

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: $N = 7,495$. Data: NDCP childcare prices, hand-collected regulation data, for 2010-2019, 2021-2022. This table shows the first stage of childcare prices on the maximum group size. “Childcare price” is the median weekly price for full-time care at a childcare centre averaged for 0 to 2 year olds, adjusted to 2010 dollars and winsorised at the 99th percentile. The Log(Price) is unadjusted. “Maximum Group Size” is the state-level maximum group size average for 0 to 2 year olds. “F-stat” is the first-stage Kleibergen-Paap Wald F statistic. All models control for county median earnings, the unemployment rate, female labour force participation, the male-female ratio, a housing price index, racial and ethnic composition, and county and year fixed effects. Estimates weighted by births to women aged 20-44. Standard errors, clustered at the state-year level, in parentheses.

We may expect the magnitude of the first stage coefficient to be larger, given that group sizes for children under the age of two are small and thus likely to be sensitive to regulatory changes. The estimates in Table 2 represent the average price response across all providers in a county, so the estimated effect will depend on the share of providers for whom the regulation is binding. The regulations are likely to bind more in some counties than others. For example, in higher-income counties, childcare providers are more likely to be operating below the regulatory maximum as parents demand higher quality childcare, so a change in the maximum could have little effect on prices at these providers, producing a dampened county-level price response.

The results in Table 3 are consistent with this. In counties with below-median household income, the relationship between the maximum group size regulation and childcare prices is stronger. A one-child increase reduces weekly childcare prices for 0 to 2-year-olds by \$5.40, roughly five times the price reduction estimated in above-median income counties (\$1.06). Interacting the instrument, however, reduces the F-statistic as the interaction now identifies heterogeneity in the first-stage relationship by income (beyond the overall relationship between maximum group size and price, which has strong predictive power overall). I therefore use the single instrument in the main analysis, and show that the second-stage results are robust to the interacted instrument in Appendix Table D3.

Table 3. First stage: interacted instrument

	Childcare Price	
	Price	Log(Price) $\times$ 100
Below-Median Income $\times$ Maximum Group Size	−4.35** (1.64)	−3.25*** (0.78)
Maximum Group Size	−1.06*** (0.22)	−0.64*** (0.13)
Mean (Childcare Price)	146.57	146.57
Mean (Maximum Group Size)	16.10	16.10
Mean (Below-Median Income, %)	36.74	36.74
Joint F-stat	4.97	7.09

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: $N = 7,495$. Data: NVSS birth records, SEER county population counts, NDCP childcare prices, hand-collected regulation data, for 2010–2019, 2021–2022. This table shows the first stage of childcare prices on a below-median-income interacted instrument. “Below-Median Income $\times$ Maximum Group Size” interacts maximum group size with an indicator for counties below the 2009 median household income. “Childcare price” is the median weekly price for full-time care at a childcare centre averaged for 0 to 2 year olds, adjusted to 2010 dollars and winsorised at the 99th percentile. “Log(Price)” is unadjusted and coefficients and standard errors are multiplied by 100. F-stat is the Kleibergen-Paap F-statistic on the excluded instruments (the interaction term and maximum group size) from the IV first stage. All models control for county median earnings, the unemployment rate, female labour force participation, the male-female ratio, a housing price index, racial and ethnic composition, and county and year fixed effects. Estimates weighted by births to women aged 20–44. Standard errors clustered at the state-year level in parentheses.

5.2 Effects of childcare prices and regulations on fertility

5.2.1 Birthrates

Table 4 shows the OLS, IV and reduced form estimates of the effects of childcare prices on birthrates, broken down by age bands. The top panel shows the OLS results, which we fear suffer from endogeneity and bias estimates towards zero due to simultaneity bias and

measurement error, as discussed in Section 4. The OLS estimates confirm this fear: they are imprecise and close to null in absolute terms.

The second panel presents the IV results estimated using the maximum group size as the instrument. The IV estimates show a statistically significant reduction in birthrates. Column 1 shows the main outcome: the log birthrate for women aged 20 to 44. A 10% increase in the weekly price of childcare (approximately \$990 a year) leads to a 5.7% decrease in the birthrate of women aged 20 to 44. Relative to the mean birthrate of 70 births per 1000 women, this amounts to 4 births per 1000. We see that this reduction in birthrates is more pronounced amongst women aged 30 and above in Columns 2 to 6; a 10% price increase has a 6 percentage point larger effect on birthrates for women aged 30 to 34 than birthrates for women aged 20 to 24. This difference appears to increase with age. The results estimated using both instruments, shown in Appendix Table D4, are consistent in coefficient size and significance.

The price elasticity estimates that I find appear large yet broadly align with magnitudes found by Mörk et al. (2013), who study the introduction of a childcare cost cap in Sweden in 2002. The authors estimate that a reduction in the present value of future marginal childcare costs of approximately \$10,000 (or \$12,200 in 2010 prices) over 10 years would increase fertility rates by 5.8%. Assuming uniform responses over time (a simplification), this corresponds to an approximate annual cost reduction of \$1,000 (or \$1,220 in 2010 prices), producing a comparable fertility response.

Another benchmark for comparison is the literature on income shocks, although price and income elasticities may not be directly comparable. Kearney and Wilson (2018) study the U.S. fracking boom between 1997 and 2012, finding that a 10% increase in earnings ($\approx$ \$7,900 in 2010 prices) leads to $\approx$ 15.8% higher birth rates (or 15.8 births per 1,000 women aged 18 to 34). This suggests a smaller income response than the price elasticity estimates I find.

The IV estimates rest on the exclusion restriction, which I discuss in Section 5.3. In the third panel, the reduced form estimates show that changes in the maximum group size regulations on their own lead to statistically significant effects on birthrates. An increase in the maximum group size average for 0 to 2-year-olds by 10 children, which could allow childcare centres to reduce prices, increases birthrates by 3.7%. Again, these effects are concentrated amongst mothers aged 30 and above.

Table 4. Effects of childcare prices on log birthrates, by age

	20-44	20-24	25-29	30-34	35-39	40-44
<i>OLS</i>						
Log(Childcare Price)	0.01 (0.02)	0.02 (0.04)	0.03 (0.03)	0.02 (0.02)	0.02 (0.02)	-0.02 (0.03)
<i>IV</i>						
Log(Childcare Price)	-0.57*** (0.14)	-0.14 (0.18)	-0.12 (0.20)	-0.73*** (0.15)	-0.87*** (0.21)	-0.80*** (0.19)
<i>Reduced Form</i>						
Maximum Group Size	0.37*** (0.13)	0.10 (0.14)	0.08 (0.13)	0.46*** (0.12)	0.54*** (0.11)	0.49*** (0.08)
Mean	69.76	97.57	122.42	89.99	37.71	7.45

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: $N = 7,495$. Data: NVSS birth records, SEER county population counts, NDCP childcare prices, hand-collected maximum group size data, for 2010-2019, 2021-2022. This table shows the estimated effects of log childcare prices on the log of birthrates per 1000 women. Reduced form coefficients and standard errors multiplied by 100. “20-44” is the log of the county-level birthrate per 1000 women aged 20 to 44, “20-24” the log of the birthrate per 1000 women aged 20-24, and so on. “Childcare price” is the median weekly price for full-time care at a childcare centre averaged for 0 to 2 year olds, winsorised at the 99th percentile. “Maximum Group Size” is the state-level maximum group size average for 0 to 2 year olds. All models control for county median earnings, the unemployment rate, female labour force participation, the male-female ratio, a housing price index, racial and ethnic composition, and county and year fixed effects. Estimates weighted by births to that age band. Standard errors, clustered at the state-year level, in parentheses.

Birthrates and preferences for non-parental childcare differ across various groups within a population, due to a range of factors such as cultural norms, career expectations, and access to childcare. Table 5 shows the effects of childcare prices on birthrates for women aged 20 to 44 decomposed by race and ethnicity. Birthrates for White, Black and Hispanic women all reduce in response to increased childcare prices. The IV estimates indicate that a 10% increase in the price of childcare reduces birthrates to White mothers aged 20 to 44 by 7.8%, to Black mothers by 5.1%, and to Hispanic women by 9%. Black mothers may be less price responsive if they are more likely to already receive public assistance. Indeed, Black children have higher rates of subsidy receipt and Head Start enrolment relative to White and Hispanic children (Hillemeier et al., 2013; Hardy et al., 2024).

Table 5. Effects of childcare prices on log birthrates, by race and ethnicity

	White, 20-44	Black, 20-44	Hispanic, 20-44
<i>OLS</i>			
Log(Childcare Price)	0.00 (0.03)	0.06 (0.05)	0.03 (0.03)
<i>IV</i>			
Log(Childcare Price)	-0.78*** (0.08)	-0.51* (0.25)	-0.90*** (0.22)
<i>Reduced Form</i>			
Maximum Group Size	0.43*** (0.11)	0.30* (0.15)	0.69*** (0.11)
Mean	66.75	58.75	81.32
	<i>White = Black</i>	<i>White = Hispanic</i>	<i>Black = Hispanic</i>
P-Value (IV)	0.032	0.473	0.051

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: $N = 7,495$. Data: NVSS birth records, SEER county population counts, NDCP childcare prices, hand-collected maximum group size data, for 2010-2019, 2021-2022. This table shows the estimated effects of log childcare prices on the log of birthrates per 1000 women, by race and ethnicity. Reduced form coefficients and standard errors multiplied by 100. “White, 20-44” is the log of the county-level birthrate per 1000 white women aged 20 to 44, and so on. “Childcare price” is the median weekly price for full-time care at a childcare centre averaged for 0 to 2 year olds, winsorised at the 99th percentile. “Maximum Group Size” is the state-level maximum group size average for 0 to 2 year olds. “P-Value (IV)” is the p-value from a Wald test of equality of the IV coefficients across race/ethnicity. All models control for county median earnings, the unemployment rate, female labour force participation, the male-female ratio, a housing price index, racial and ethnic composition, and county and year fixed effects. Estimates weighted by births to that race-age group. Standard errors, clustered at the state-year level, in parentheses.

5.2.2 Birth timing

Next, I explore whether birth timing responds to changes in childcare prices using the individual-level sample of births. Table 6 displays the OLS, IV, and reduced form results for mother’s age at first birth and spacing between births. I caveat this analysis by noting that these estimates also reflect compositional change in the sample of mothers. Mothers who choose to not have a first, second, or third child in response to price increases may differ in unobservable characteristics to those who remain in the sample, and so these estimates should be viewed as descriptive and suggestive of behavioural responses.

I find that a rise in childcare prices is associated with mothers shifting their first birth into the future. The IV estimates reveal that a 10% increase in the weekly price of centre-based care is associated with age at first birth rising by 0.3 years, or roughly 4 months. I also

find evidence of an increase in time between the first and second birth; a 10% increase in the weekly price of childcare increases the time between the first and second birth by 0.4 months. There is a negative statistically significant effect on time between the second and third birth, which is the opposite direction to the prediction that higher childcare prices may lead parents to delay births. This effect, however, could be explained by mothers who would otherwise have had longer birth intervals deciding not to have a third child. In addition, the mothers who do choose to have a third birth may be more likely to have shorter birth intervals and also live in high price areas.

The reduced form estimates also show a statistically significant effect of maximum group size regulations on age at first birth and spacing between the first and second birth. An increase in the maximum group size by 10 children is associated with a decrease in the age of first time mothers of 0.19 years (2.2 months), and a decrease in the first birth interval by 0.26 months. We see an effect in the opposite direction for the interval between the second and third birth; the discussion in the previous paragraph on sample compositional change applies here too.

Table 6. Effects of childcare prices on birth spacing outcomes

	Age at 1st birth		Birth spacing (1-2)		Birth spacing (2-3)	
<i>OLS</i>						
Log(Childcare Price)	-0.16 (0.11)	-0.14 (0.10)	-0.02 (0.19)	-0.08 (0.14)	0.38 (0.50)	0.34 (0.52)
<i>IV</i>						
Log(Childcare Price)	1.72** (0.77)	3.34*** (0.52)	6.20*** (0.33)	4.43*** (0.54)	-7.29** (2.88)	-8.46*** (2.87)
<i>Reduced Form</i>						
Maximum Group Size	-0.95** (0.47)	-1.85*** (0.44)	-3.60*** (0.58)	-2.57*** (0.87)	4.48** (1.82)	5.19*** (1.62)
Mean	27.60	27.60	50.52	50.52	52.71	52.71
Controls	Cty	Cty+Ind	Cty	Cty+Ind	Cty	Cty+Ind
N	4,178,730	4,178,730	3,418,884	3,418,884	1,864,605	1,864,605

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Data: NVSS birth records, NDCP childcare prices, hand-collected maximum group size data, for 2010-2019, 2021-2022. This table shows the estimated effects of log childcare prices on birth spacing outcomes. Reduced form coefficients and standard errors multiplied by 100. “Age at 1st birth” is the age at which the mother has their first child in years. “Birth spacing (1-2)” is the number of months between the first and second birth. “Birth spacing (2-3)” is the number of months between the second and third birth. “Childcare price” is the median weekly price for full-time care at a childcare centre averaged for 0 to 2 year olds, winsorised at the 99th percentile. “Maximum Group Size” is the state-level maximum group size average for 0 to 2 year olds. “Cty” controls include county median earnings, the unemployment rate, female labour force participation, the male-female ratio, a housing price index, the mother’s race and ethnicity, and county and year fixed effects. “Cty&Ind” controls also include the mother’s date of birth and marital status. Standard errors, clustered at the state-year level, in parentheses.

5.3 Threats to identification

5.3.1 Interpretation of childcare price variable

In the above analysis I use the price of centre-based childcare as a proxy for the price of childcare more broadly, supported by two analyses. First, I show that childcare centre prices are representative of the formal care market. The NDCP data also contain prices for family-childcare homes for a reduced sample. Appendix Figure E2 shows that centre and family-based care prices in the NDCP are highly and statistically significantly correlated. Second, I examine the price of informal childcare using nannies' wages in the Current Population Survey (CPS).⁹ Appendix Table D5 shows the effects of changes in childcare centre regulations on weekly wages of nannies. Stricter regulations (reflected in smaller maximum group sizes) increase nannies' wages, consistent with families substituting to informal childcare. Changes in the staff-child ratio yield imprecisely estimated coefficients. That stricter regulations increase both centre-based prices and nannies' wages suggests that formal and informal childcare are close substitutes, supporting the use of centre-based prices as a proxy for broader childcare market prices.

There may be mismeasurement of childcare prices at the county level for states which report childcare prices at the market, rather than county, level. Childcare markets, or "market rate regions" (MRR) henceforth, are used by states for CCDF childcare subsidy rate setting. For example, Oregon set their regions by examining the variation in prices across the state (Grobe et al., 2003). States also aggregate data across counties to overcome identification concerns, as some counties will have a small number of childcare establishments. If market rate regions closely correspond to the geographic markets in which families search for childcare, then the market rate region prices provide a reasonable measure of the price faced by families in a county. If households face prices that differ from the market rate region price, then measurement error is present. Measurement error is minimised when states group counties based on similar underlying economic and demographic characteristics, or relatively homogeneous true prices. Under classical measurement error, IV estimates identify the effect of the (unobserved) true county-level price on birthrates, as state-level regulations are uncorrelated with the measurement error. If within-MRR price deviations are correlated with unobserved determinants of birthrates, non-classical measurement error is present. In Appendix Table D6, for the counties that are grouped into market rate regions, I show the

⁹Detail on sample construction and methods are contained in Appendix C.

mean, standard deviation, and coefficient of variation within a market rate region for various characteristics. Counties are grouped with similar income and labour market conditions, plausibly the dominant drivers of childcare prices on the demand and supply side. There is more within-MRR heterogeneity in racial and ethnic composition. In this case, the IV estimate is better interpreted as the effect of market-rate-region price variation on birthrates. Note, however, that because the instrument varies at the state level, having data structured at the market rate region does not lead us to lose any first-stage identifying variation.

5.3.2 Exclusion restriction

One may worry that childcare regulations may impact not only the price of childcare, but also the viability of childcare businesses. A reduction in the number of providers could, in turn, influence fertility decisions. For example, whether the closure of a local provider leads families to reconsider having a child. In Appendix Table D7, I present evidence that mitigates this concern; my estimates are robust to controlling for the share of childcare establishments per child aged 0 to 5.

In addition, childcare regulations not only affect the price of childcare, but also the quality. If potential parents care sufficiently about childcare quality, their fertility choices may respond to changes in quality. If such a channel exists, my estimates of the effect of the price of childcare on birthrates may be biased. One approach to mitigate this concern is to control for a measure of childcare quality. Obtaining data on childcare quality is not trivial as there is a dearth of publicly available, nationally representative data on childcare quality.¹⁰

I can, however, control for one measure of childcare quality: staff turnover. Research in the early-childhood literature indicates that staff turnover in the childcare centre can create instability in the care environment, disrupting relationships (Howes and Hamilton, 1992; Doromal et al., 2022). Stable relationships are important for child development (Lally and Mangione, 2017). In addition, directors' attention may get directed away from the children towards hiring, and until open jobs are filled, remaining staff may need to adjust their activities, take on more children, or work overtime (Whitebook and Sakai, 2004). Each of these factors could lead to reduced quality in the provision of care. In Appendix Table D8 I control for the staff turnover rate in the childcare sector; the estimates are robust to this

¹⁰A system of childcare quality rating does exist in the U.S. - the Quality Rating and Improvement System (QRIS). However, QRIS systems are designed and monitored at the state level, so vary across states, and not all states have a QRIS system. Furthermore, states do not publish their historical QRIS data.

inclusion.

Nonetheless, I proceed to quantify the effect of the quality channel. To do so, I conduct a bounding exercise in Appendix A to explore how such a quality channel affects the size and direction of my estimates. To quantify the potential bias, I use data for a subset of sample years on a measure of childcare quality - childcare programme accreditation - from the National Association for the Education of Young Children (NAEYC) ([National Association for the Education of Young Children, 2025](#)). As the NAEYC was unable to share data prior to 2017, I cannot use these data as a control variable in robustness analysis. First, I show that my IV estimates represent a lower-bound of the effect of childcare prices on birthrates. Next, using the NAEYC data, I estimate the magnitude of the true β under the assumption that quality of childcare directly impacts birthrates. I describe the NAEYC data and set out the bounding analysis in Appendix A.

5.3.3 Additional assumptions

The instrument must also address conditional independence. Conditional on covariates and fixed effects, the childcare regulations must be as good as randomly assigned with respect to fertility outcomes. Including county fixed effects absorbs time-invariant local factors (e.g. cultural norms), whilst year fixed effects account for national level shocks or trends. To further strengthen the credibility of conditional independence, I include a set of time-varying control variables that could affect family formation. I control for county-year changes in socioeconomic factors such as earnings and the unemployment rate, racial and ethnic composition, and house prices. These controls help address potential confounding from within-county trends that may be correlated with the regulations and birthrates.

In the presence of heterogeneous treatment effects, to interpret the instrumental variables (IV) estimates as the LATE, we also require the monotonicity assumption. In this setting, the monotonicity assumption demands that a decrease in the maximum group size weakly increases or weakly decreases childcare prices for all counties. Given that a decrease in the maximum group size will reduce profits for childcare providers, I expect such a change to increase childcare prices for all. It seems implausible that the regulations would have the opposite effect. Nonetheless, to assess this assumption, I conduct one-sided Kolmogorov-Smirnov tests comparing childcare price distributions in treated and untreated county-years. The null hypothesis is that the prices for treated county-years come from the same distri-

bution as prices for untreated county-years. The alternative hypothesis is that the price distribution of treated county-years stochastically dominates that of untreated county-years. The null hypothesis is rejected with $p < 0.001$. A second Kolmogorov-Smirnov test reversing the direction of the alternative fails to reject the null. Together, these results give additional reassurance that the monotonicity assumption is reasonable.

The Stable Unit Treatment Value Assumption (SUTVA) states that the potential outcome for one unit must be unaffected by the assignment of treatment to the other units. In my setting, this implies that birthrates in a given county are unaffected by changes in maximum group size or staff-child ratio regulations in other counties. Where we may have concerns about this assumption is at state borders. If childcare regulations shift the price of childcare within a state, families in counties bordering a state with lower childcare prices may choose to send their children to childcare providers across the state border. Such cross-border substitution would violate SUTVA. To address this concern, I drop bordering counties from my sample. As shown in Appendix Table D9, the magnitude and significance of the estimates remain unchanged.

Given that I have a small number of “treated” states (states which experience changes in the regulations in my sample) one may worry that there is effect heterogeneity and that the LATE does not closely approximate the Average Treatment Effect. The LATE that I estimate is the weighted causal effect on women in counties in the treated states. In Section 5.4, I show that the results are not sensitive to dropping states one at a time, which provides some assurance that effect size heterogeneity is not a substantial issue. One may also question whether the treated and control states were on different paths. In Appendix Figures E4, E5, and E6, I plot the event study estimates of the effect of the introduction of childcare facility regulations and birthrates for women aged 30 to 34 (a group driving the results) for each of the three treated states.¹¹ For Nevada and Delaware, the regulations introduced were more stringent on childcare centres, and in these two states we see no evidence of pre-trends and evidence of reductions in birthrates. In Vermont, the state relaxed the group size requirement for 18 month olds, so we may expect to see increases in birthrates. The estimates are noisy, but point in this direction. Overall, these event studies provide some reassurance that counties in control states in combination with covariates are providing a suitable control for counties in treated states.

¹¹Note that this is not exactly the same approach as the reduced form analysis I present in Section 5, where I use the continuous variables of the maximum group size (e.g. 15) and staff-child ratio (e.g. 0.25) rather than a binary indicator for presence of a policy.

Moreover, given declining rates of fertility in the U.S., one might be concerned that states may relax these regulations to encourage fertility. Yet two of the three regulatory changes that I observe increase the stringency of the mandates on childcare providers (e.g. introducing a maximum group size requirement). Additionally, reporting and discussion of the regulations highlights the key goal of these regulations: to protect child health and safety and encourage learning (See Appendix Figure E3). Mandated group size and staff-child ratio requirements support child safety. They ensure staff can adequately supervise and care for the children they are responsible for, reducing the risk of injuries or accidents. The requirements also support child development. Sufficient room space limits overcrowding and promotes increased staff-child and safe peer interactions, which can enhance learning and development.

5.4 Robustness

I explore the sensitivity of my findings to various specifications and approaches. Overall, the estimates and quantitative conclusions remain stable. First, the results are robust to the inclusion of both instruments, rather than just the maximum group size instrument. Appendix Table D2 shows the first stage using both instruments, for both the main sample (that exploits changes in NV and VT) and a robustness sample (that exploits changes in NV, VT, and DE). We see that the staff-child ratio is a strong predictor of prices, with a large F-statistic on this first stage. Furthermore, both of these specification-sample combinations deliver similar point estimates and significance in the second stage as the main specification (see Appendix Table D4). The IV estimates are also robust to weighting by population rather than number of births (See Appendix Table D10). Next, in Appendix Table D11, I repeat the IV analysis of the effects of childcare prices on birthrates for women aged 20 to 44 on subsamples of states, dropping one state at a time. This exercise demonstrates that the results are not driven by any one state; they are consistent and statistically significant across all subsamples. My main analytical approach winsorises childcare prices at the 99% level as there are notable outliers, as shown in Appendix Figure E7. Appendix Table D12 illustrates that my main results are robust to different levels of winsorising of childcare prices, and not winsorising at all. The IV estimates are slightly larger when I don't winsorise prices, but once the influence of the outliers is reduced, there is little difference in coefficient values from winsorising at the 95th, 98th, or 99th percentile. My primary approach is to cluster standard errors at the state-year level, given concerns for the bias that can result from having too few clusters (clustering at the state level delivers 30 clusters). Appendix Table D1 shows that the results withstand a more stringent approach to clustering. The estimates remain

significant at the 1 percent level if I cluster standard errors at the state level. I also test the sensitivity of my estimates to alternative ways of handling states without maximum group size regulations. Some states have no maximum group size restrictions in place, including two states which experience changes in the regulations across the study period. In order to retain observations in these states in my analytical sample, I set these missing values to 30 in my baseline specification. This number seems reasonable given that the maximum value in my data is 22. Appendix Table D13 demonstrates that the estimates are not sensitive to setting these missing values to alternative numbers. Finally, the estimates are robust to excluding control variables that may themselves be outcomes. In Table D14, I replace the unemployment rate and median earnings with their male-specific counterparts and remove female labour force participation as a covariate.

6 A model of fertility choices

In this section, I use a theoretical model of household decision making to outline mechanisms behind the age gradient in price elasticities. The model is built on work by Doepke et al. (2023); I allow for part-time childcare and build in human capital accumulation with two periods to consider age heterogeneity. Most detail can be found in Appendix B; here I set up the model and outline the key model predictions.

6.1 Setup

I model the household from the perspective of an individual who gives birth, who has a working partner, and whom I assume represents the views of the household. Henceforth, I will assume that this individual is a woman to align with the birth records analysis. In this model, the woman derives utility from consumption and having children, and either a benefit or dis-utility from caring for her child(ren) at home. There are two periods, $t \in \{1, 2\}$. In each period, a woman decides whether she will have a child. She can have at most one child each period: $n \in \{0, 1\}$. The preferences of the women are described by the following utility function:

$$u = c_1 + c_2 + (2n_1 + n_2)v - \mu((1 - s_1)^\gamma n_1 + (1 - s_2)^\gamma n_2)$$

Each period lasts for four years, such that childcare is only required for the child born in that period. This setup allows us to consider fertility responses of younger and older mothers.

The mother's budget constraints for periods one and two are given by:

$$c_1 + ps_1n_1 \leq w(1 - (1 - s_1)n_1)$$

$$c_2 + ps_2n_2 \leq w(1 + \alpha h_1)(1 - (1 - s_2)n_2)$$

Each mother is endowed with one unit of time, which can be allocated between employment and childcare (s). The mother can choose to care for the child at home, or purchase childcare in the private market at price p per unit of time. She selects the desired share of private childcare, s , in each period. If the woman works, she earns a wage w in $t = 1$. Working in the first period allows the woman to build human capital h_1 , which delivers a wage increase in the second period. Her period $t = 2$ wage is $w(1 + \alpha h_1)$. The parameter α captures productivity.

Women can experience either a positive benefit or some dis-utility from caring for their child at home. I initially present the utility function with a dis-utility from self-care, but in subsection B.4 I model a positive benefit of self-care and my key model predictions are unchanged. The dis-utility term puts a greater penalty on longer periods of self-childcare at home ($\gamma > 1$). This term reflects a documented preference for working rather than home-making (Gallup, 2019). Note that this does not mean that the woman does not enjoy any child-rearing, only that she has a preference for some working hours, rather than caring for her child at home full-time.

In this simple model, I assume that the woman has a partner with exogenous income who can support the household if there is no outsourcing of childcare, that she represents the views of the household, and that there is no time discounting. I abstract from savings.

6.2 Model predictions

At first glance, it is unclear whether younger or older women would be more sensitive to changes in the price of childcare. One on hand, younger women are more likely to earn less than their older counterparts, which could make them disproportionately affected by price hikes. Yet there are mechanisms that could drive greater price responsiveness amongst

older women. For example, older mothers may be on their second or third birth and higher parity births may be more price reactive. Additionally, the higher wages and developed careers of older women raise the opportunity cost of having a child. My model explores these mechanisms for older women and delivers the following propositions:

Proposition 1. *Women who earn higher wages will outsource more, if not all, childcare and remain in the labour market.*

Higher wages increase the opportunity cost of time, raising the incentive to outsource more childcare. In comparison, women with lower wages will provide more self-childcare at home and decrease their labour supply. In this way, women with lower wages will be less exposed to childcare prices and in turn less reactive to price changes. Formally, the first order condition for the optimal private childcare share s^* yields

$$\frac{\partial s^*}{\partial w} > 0,$$

so that as w increases, s^* rises and exposure to childcare prices grows. This result can be found in Appendix B.

Proposition 2. *An increase in the childcare price reduces the probability of (older) mothers having a birth in the second period.*

Amongst mothers with one child, price increases decrease the probability of a second-period birth and raise the probability of a first period birth. As work experience builds human capital, second-period wages are higher. This leads older mothers to purchase a larger share of private childcare than mothers who give birth earlier. Thus price increases will hurt older mothers with greater reliance on paid childcare more than their younger counterparts. Formally,

$$\frac{dP_{01|1}}{dp} < 0$$

This result is derived in Appendix B.

Proposition 3. *Second births are more price sensitive than first births, as total childcare spending rises with additional children.*

A mother with two children is exposed to paid childcare in both periods, so total spending increases (if a mother outsources some share of private childcare in the first period, she will also outsource in the second period). This effect is amplified by the fact that due to the wage uplift in period two, private childcare is more attractive in the second period. Formally, the difference in the elasticities between having two children and having one child is

$$\eta_{11,p} - \eta_{10,p} = -p s_2^* < 0$$

which shows that the marginal impact of a price increase is larger for higher parity births. The proof can be found in [Appendix B](#).

7 Mechanisms

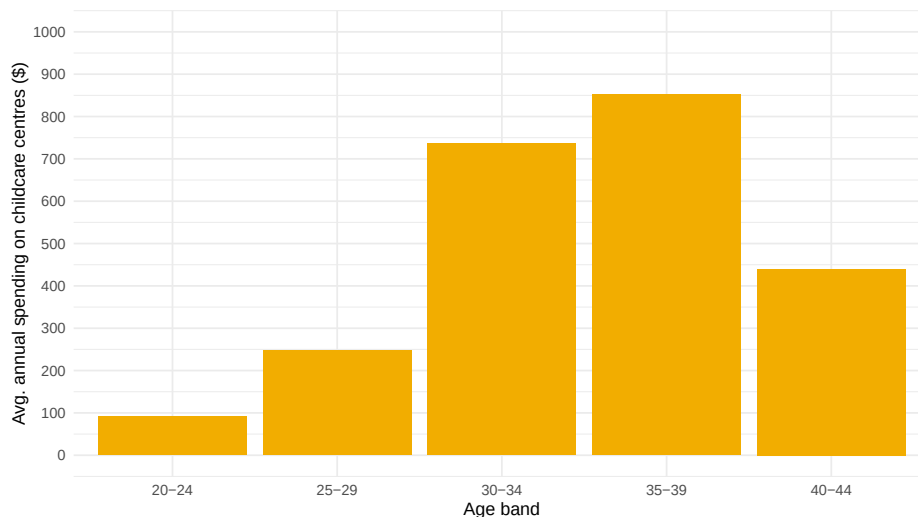
Next, I test the model predictions for why older women are more childcare price responsive than younger women. I show that older women allocate more spending to formal childcare than younger women. I provide evidence that higher-income women are more price responsive, and that within income groups this responsiveness increases with age. I find suggestive evidence that higher parity births exhibit greater responsiveness to childcare prices. Together, these mechanisms help to explain the age gradient in price sensitivity.

7.1 Older parents spend more on childcare

My model predicts that older women with higher incomes will outsource more childcare. This could explain the age gradient in the price elasticity of fertility, if the price responsiveness of older mothers reflects their greater reliance on formal care. Younger mothers may rely more on informal care, perhaps by parents or relatives, or may be more likely to drop out of the labour market to care for the child themselves. Indeed, I do not find statistically significant birthrate responses amongst younger women.

I explore whether older mothers use more formal childcare by examining spending on childcare centres in the Consumer Expenditure Survey. Figure 5 shows the average annual spending on childcare centres by age in 2022. Appendix Figure E8 shows the same plot for 2010. We observe a noticeable pattern by age; spending on childcare centres remains low for parents in their early to mid twenties, rises sharply for those in their thirties, before falling for those in their forties. The age gradient is seen in both 2010 and 2022, but there is a more pronounced increase in spending for parents in their thirties in 2022.

Figure 5. Avg. annual spending on childcare centres for 2022, by age



Note: Data: Consumer Expenditure Survey, for 2022. This figure shows the average annual spending on childcare centres by age of the respondent, for respondents with any children under 3, adjusted for inflation using base year 2010. Childcare centres defined as day care centres, nurseries, and preschools.

7.2 Higher educated women (a proxy for income) are more price sensitive

A second prediction is that older women with higher incomes, through outsourcing more childcare, will be more price sensitive. The birth records data does not contain mother's income, but I observe education, which I use as a proxy for income.¹² I find that women with an undergraduate/Bachelor's degree are more responsive to childcare price changes than women without a Bachelor's degree.¹³ Furthermore, within education, the coefficient sizes

¹²Note that due to exclusion of the education variable for many states prior to 2014 by NVSS, I run this heterogeneity analysis on a subsample of the following years.

¹³Note that the 5-year ACS only provides population counts by age and education for women aged 18 to 24, and then by 10 year age bands, hence why the age bands in Table 7 differ from other analyses in this paper.

increase with age, further supporting the conclusions from my model. We also observe these patterns in the reduced form.

Table 7. Effects of childcare prices on log birthrates, by education

	BA			No BA		
	18-44	25-34	35-44	18-44	25-34	35-44
<i>OLS</i>						
Log (Childcare Price)	-0.00 (0.04)	0.00 (0.05)	-0.02 (0.03)	0.00 (0.04)	0.03 (0.05)	-0.04 (0.05)
<i>IV</i>						
Log (Childcare Price)	-1.94*** (0.02)	-2.05*** (0.20)	-3.47*** (0.19)	-1.20** (0.38)	-1.27*** (0.13)	-2.70** (1.01)
<i>Reduced Form</i>						
Maximum Group Size	0.37*** (0.10)	0.45** (0.15)	0.42** (0.16)	0.32* (0.14)	0.33* (0.16)	0.53** (0.17)
Mean	71.83	127.60	32.10	63.93	99.48	19.63
	<i>BA = No BA, 18-44</i>		<i>BA = No BA, 25-34</i>	<i>BA = No BA, 35-44</i>		
P-Value (IV)	0.03		0.18	0.07		

* p < 0.10, ** p < 0.05, *** p < 0.01.

Notes: N = 4,823. Data: NVSS birth records, ACS population by education counts, NDCP childcare prices, hand-collected maximum group size data, for 2010-2019, 2021-2022. This table shows the estimated effects of log childcare prices on the log of birthrates per 1000 women, by education. Reduced form coefficients and standard errors multiplied by 100. “18-44” is the log of the county-level birthrate per 1000 women aged 18 to 44, “25-34” the log of the birthrate per 1000 women aged 25-34, and so on. “BA” is women with a Bachelor’s degree. “No BA” is women with less than a Bachelor’s degree. “Childcare price” is the median weekly price for full-time care at a childcare centre averaged for 0 to 2 year olds, winsorised at the 99th percentile. “Maximum Group Size” is the state-level maximum group size average for 0 to 2 year olds. “P-Value (IV)” is the p-value from a Wald test of equality of the IV coefficients across education. All models control for county median earnings, the unemployment rate, female labour force participation, the male-female ratio, a housing price index, racial and ethnic composition, and county and year fixed effects. Estimates weighted by births to that age and education band. Standard errors, clustered at the state-year level, in parentheses.

7.3 Higher order birthrates are more price sensitive

A third prediction is that mothers with more children are more price sensitive because total childcare spending rises. Older mothers are more likely to be having their second or third child than younger mothers. I test this mechanism through heterogeneity analysis by parity.

Table 8 presents estimates of the effect of childcare prices on first, second, and third birthrates. Point estimates suggest that, in percentage terms, second and third birthrates are more responsive to childcare prices than first birthrates among women aged 20 to 44. For a 10%

increase in the weekly price of childcare, there is a fall in first birthrates for women aged 20 to 44 by 2.4%. This is a decline in 0.5 births per 1000 relative to the mean of 22.7. For second birthrates, this figure is 4.3% (0.9 births per 1000) and for third birthrates, 3.9% (0.5 births per 1000). However, the differences in the estimated effects across parity are not statistically significant and thus should be interpreted as suggestive.

Table 8. Effects of childcare prices on log birthrates, by parity

	First	Second	Third
Log (Childcare Price)	-0.24*** (0.09)	-0.43*** (0.15)	-0.39** (0.15)
Mean	22.66	20.64	12.09
	<i>First = Second</i>	<i>First = Third</i>	<i>Second = Third</i>
P-Value	0.22	0.46	0.65

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: $N = 7,495$. Data: NVSS birth records, SEER county population counts, NDCP childcare prices, hand-collected maximum group size data, for 2010-2019, 2021-2022. This table shows the estimated effects of log childcare prices on the log of first, second, and third birthrates per 1000 women aged 20 to 44. "Childcare price" is the median weekly price for full-time care at a childcare centre averaged for 0 to 2 year olds, winsorised at the 99th percentile. "P-Value" is the p-value from a Wald test of equality of the IV coefficients across parity. All models control for county median earnings, the unemployment rate, female labour force participation, the male-female ratio, a housing price index, racial and ethnic composition, and county and year fixed effects. Estimates weighted by total births to that parity band. Standard errors, clustered at the state-year level, in parentheses.

8 Conclusion

Low and falling fertility rates across the developed world are generating concern about future economic growth and the financial viability of social support systems. In this paper I explore how the price of childcare, a large and early cost facing potential parents, affects the decision to have children in the U.S. I provide the first causal evidence on how changes in childcare prices affect fertility behaviour, and extend a limited reduced form literature exploring the causal effects of childcare costs on fertility.

I find that higher childcare prices reduce birthrates, delay the first birth, and increase spacing between the first and second births. Heterogeneity analysis reveals varying effects by age: the price elasticity for women aged 30 and above is larger than that for younger women. I develop a simple theoretical model to explain this age gradient. I demonstrate that older women can be more price responsive than younger women because they earn a higher wage, and so the opportunity cost of their time is higher. This higher opportunity cost leads them

to outsource more, if not all, childcare. This high level of outsourcing makes women more sensitive to price changes. Older women are also more likely to be having a second, or higher, birth. Having additional children increases spending on childcare, increasing price sensitivity. Using data I demonstrate that, indeed, higher-income women (proxied by education) are more price responsive, and that older parents spend more on formal childcare. Additionally, I find suggestive evidence that price elasticities are larger for higher-order births, indicative of stronger responses from women along the intensive margin.

These results suggest that preventing rising childcare costs and expanding financial support for childcare are promising avenues for policymakers seeking to address low fertility rates in the U.S. In particular, sustaining birthrates among women aged 30 and above is critical for any recovery in completed fertility, as current cohorts of women will need to have more births after age 30 to match total fertility rates of cohorts prior to 2007 ([Kearney and Levine, 2021](#)).¹⁴

To illustrate the potential policy implications of my estimates, one can conduct a simple and illustrative back-of-the-envelope calculation of the cost and fertility effects of a childcare price subsidy for parents of children under age three. Using a childcare centre participation rate of 19.61%¹⁵ and a population estimate of 11 million children under 3 in 2023, I estimate that 2.16 million children are enrolled in centre-based care. The weighted mean weekly price of childcare is \$190 in my sample, or \$9,880 annually. A 10% annual subsidy would therefore cost approximately \$2.13 billion. Applying my estimated birthrate elasticity (which assumes that this is representative of the national population), such a subsidy would raise birthrates by 5.7%, implying approximately 205,000 additional births in the first year.¹⁶ This implies a cost of approximately \$10,400 per additional birth in the first year.

¹⁴The total fertility rate is the number of children a woman is expected to have over her childbearing years, given current age-specific fertility rates.

¹⁵53% of children under 3 are in non-parental care, 37% of these are in centre care ([National Center for Education Statistics, 2020](#)).

¹⁶Relative to a baseline of 3,596,017 births in 2023.

A Appendix A: Bounding exercise

In this section, I conduct a bounding exercise to explore how the quality of childcare may affect my IV estimates. I show that my IV estimates represent a lower-bound, and estimate the size of the bias in the case where quality of childcare directly impacts birthrates.

A.1 Decomposition of the IV Estimator

In the two stage-least squares equation I set out in section 4, I assume that:

$$Y = \beta P + X'\eta + \epsilon \quad (5)$$

Where Y is the log of the birthrate, P is the cost of childcare, and X' contains covariates including county and year fixed effects. As P is endogenous, I instrument the variable with Z :

$$P = \mu Z + X'\kappa + \varepsilon \quad (6)$$

However, let us consider the possibility that the structural model is in fact:

$$Y = \beta P + \delta Q + X'\eta + e \quad (7)$$

Where Q is the quality of childcare, and the regulations that I use as a instrument for P also affect Q . If we estimate equation 7 using equation 5, then the error term may include the quality of childcare component:

$$\epsilon = \delta Q + e$$

By the Frisch-Waugh-Lovell theorem, I partial out the observed covariates X :

$$\begin{aligned} \tilde{Y} &= \beta \tilde{P} + \delta \tilde{Q} + \tilde{e}, \\ \tilde{P} &= \mu \tilde{Z} + \tilde{\varepsilon} \end{aligned}$$

Then the instrumental variables estimator of β can be written as:

$$\begin{aligned}\hat{\beta}_{IV} &= \frac{\text{Cov}(\tilde{Z}, \tilde{Y})}{\text{Cov}(\tilde{Z}, \tilde{P})} \\ &= \frac{\text{Cov}(\tilde{Z}, \beta \tilde{P} + \delta \tilde{Q} + \tilde{e})}{\text{Cov}(\tilde{Z}, \tilde{P})} \\ &= \frac{\beta \text{Cov}(\tilde{Z}, \tilde{P}) + \delta \text{Cov}(\tilde{Z}, \tilde{Q}) + \text{Cov}(\tilde{Z}, \tilde{e})}{\text{Cov}(\tilde{Z}, \tilde{P})}.\end{aligned}$$

With the assumption that $\text{Cov}(\tilde{Z}, \tilde{e}) = 0$,

$$\hat{\beta}_{IV} \rightarrow \beta + \delta \cdot \frac{\text{Cov}(\tilde{Z}, \tilde{Q})}{\text{Cov}(\tilde{Z}, \tilde{P})}$$

Therefore, to bound $\hat{\beta}_{IV}$, we must estimate $\text{Cov}(\tilde{Z}, \tilde{Q})$ and $\text{Cov}(\tilde{Z}, \tilde{P})$. We can estimate $\text{Cov}(\tilde{Z}, \tilde{P})$ using the first stage, equation 6. If:

$$\tilde{P} = \mu \tilde{Z} + \tilde{\varepsilon}$$

It follows that:

$$\hat{\mu} = \frac{\text{Cov}(\tilde{Z}, \tilde{P})}{\text{Var}(\tilde{Z})}$$

$$\Rightarrow \text{Cov}(\tilde{Z}, \tilde{P}) = \hat{\mu} \cdot \text{Var}(\tilde{Z})$$

Say the regulations (Z) also affect the quality of childcare (Q), then we can consider the following equation to estimate $\text{Cov}(\tilde{Z}, \tilde{Q})$:

$$\tilde{Q} = \omega \tilde{Z} + \tilde{u} \tag{8}$$

It follows that:

$$\hat{\omega} = \frac{\text{Cov}(\tilde{Z}, \tilde{Q})}{\text{Var}(\tilde{Z})}$$

$$\Rightarrow \text{Cov}(\tilde{Z}, \tilde{Q}) = \hat{\omega} \cdot \text{Var}(\tilde{Z})$$

Substituting in these expressions for $\text{Cov}(\tilde{Z}, \tilde{Q})$ and $\text{Cov}(\tilde{Z}, \tilde{P})$, we can bound the estimate of β using the following equation:

$$\hat{\beta}_{IV} - \hat{\delta} \cdot \frac{\hat{\omega}}{\hat{\mu}} \rightarrow \beta$$

Where $\hat{\delta}$ is an estimate of the effect of childcare quality on birthrates, $\hat{\omega}$ is an estimate of the effect of the instrument (childcare regulations) on childcare quality, and $\hat{\mu}$ is an estimate of the effect of the instrument on childcare prices.

A.2 Signing the bias

Let us first consider the sign of this bias. I find that an increase in childcare prices reduces birthrates, so the estimator $\hat{\beta}_{IV} < 0$. For the main instrument, the maximum group size, I find that an increase in the maximum group size reduces prices, so $\hat{\mu} < 0$. We would expect that loosening the regulations, by increasing the maximum group size, could reduce the quality of childcare. This would leave $\hat{\omega} < 0$. Finally, if childcare quality were to affect birthrates, we would anticipate that raising quality would have a positive effect on birthrates ($\hat{\delta} > 0$). Putting these together, we get:

$$\hat{\beta}_{IV} \underset{(<0)}{=} \underset{(>0)}{\hat{\delta}} \cdot \frac{\underset{(<0)}{\hat{\omega}}}{\underset{(<0)}{\hat{\mu}}} \longrightarrow \underset{(<0)}{\beta}$$

If we instead consider the minimum staff-child ratio, an increase in the staff-child ratio increases prices, so $\hat{\mu} > 0$. An increase in the staff-child ratio makes the regulations stricter, so we would expect this to increase the quality of childcare. Thus $\hat{\omega} > 0$. As before, we would anticipate that raising quality has a positive effect on birthrates; $\hat{\delta} > 0$. Combining these we get:

$$\hat{\beta}_{IV} \underset{(<0)}{=} \underset{(>0)}{\hat{\delta}} \cdot \frac{\underset{(>0)}{\hat{\omega}}}{\underset{(>0)}{\hat{\mu}}} \longrightarrow \underset{(<0)}{\beta}$$

This analysis reveals that if quality of childcare is a pertinent omitted variable, the estimator $\hat{\beta}_{IV}$ is an underestimate of the true value of β . Next, I will estimate the magnitude of the bias term.

A.3 Estimating the magnitude of the bias

Next, I estimate the effect of childcare quality on birthrates ($\hat{\delta}$) and the effect of childcare regulations on childcare quality ($\hat{\omega}$). Here, the measure of childcare quality that I use is the fraction of childcare establishments accredited by the National Association for the Education of Young Children (NAEYC) ([National Association for the Education of Young Children, 2025](#)).

The NAEYC has been an independent accreditor of early learning programmes for over thirty years, working with facilities across the U.S. to improve quality and implement best practices. NAEYC accreditation provides a signal to families that the childcare programme is reputable and high quality. Early learning programmes, including childcare centres, apply for NAEYC accreditation voluntarily. During my sample period, there were four stages of the NAEYC early learning accreditation process ([National Association for the Education of Young Children, 2023](#)). First, a programme will enrol and undertake self-study in preparation. Next, they will apply for accreditation and conduct a self assessment to demonstrate that they meet the NAEYC standards. If the programme has sufficiently evidenced that they are ready to progress to the next stage, they will receive a site visit by a NAEYC assessor. During the site visit, programmes are marked against several categories of quality: curriculum, teaching, relationships with children, families, and communities, child learning and development, health and safety, staff competencies, the physical environment, programme management, and collaboration with communities. Accredited programmes are awarded accreditation for five years (subject to meeting annual maintenance requirements). I have data on programme accreditation for 2017 to 2022 inclusive (the NAEYC was unable to share data prior to 2017). The lack of earlier data is a limitation of my analysis, but the remaining sample still allows me to estimate the effect of childcare quality on birthrates for the purposes of this bounding exercise.

With these data, I estimate the fraction of childcare establishments that are NAEYC accredited at the state level. To do so, I divide the number of NAEYC accredited programmes by the number of childcare establishments in the Census CBP data at the state level for each of the years from 2017 to 2022 inclusive. Across this time period and sample, the average share of childcare establishments that are NAEYC accredited is 8.4%. A caveat is in order. I may underestimate the number of accredited programmes prior to 2022, as the NAEYC were unable to share information on programmes that lost accreditation status. However, I do not see a large jump in the share of accredited programmes in 2022 when I plot the mean share

over time (See Appendix Figure E9). Further, to test the sensitivity of my estimates, I drop states where I observe a noticeable rise in the number of programmes in 2022. Reassuringly, the coefficients' significance, direction, and rough magnitude remain consistent.

First, I estimate the effect of childcare quality on birthrates ($\hat{\delta}$). I present OLS estimates of the effect of childcare quality on birthrates in Table A1. The share of NAEYC programmes has an insignificant positive effect on log birthrates.

Next, I estimate the effect of the childcare regulations on childcare quality ($\hat{\omega}$). For this component, I have two approaches. First, I estimate the effect of maximum group sizes/staff-child ratio changes on the NAEYC accreditation rate using my hand-collected data on childcare facility regulations. Across the years 2017 to 2022 there were two changes: Virginia introduced group size requirements (2021), and Utah relaxed their group size requirements (2022). As I only have pre- and post- years for Virginia, I exploit the change in Virginia. The results are shown in Table A2. I find that increasing the maximum group size has a significant but small effect on the share of NAEYC accredited programmes. For the second approach, I look to prior work by Hotz and Xiao (2011). The authors estimate the effect of the minimum staff-child ratio for infants on the NAEYC accreditation rate; they estimate a significant positive coefficient of 0.639.¹⁷

¹⁷Please see “Table 12 - Estimated Effects of State Regulations on the Accreditation of Child Care Centers” on page 1802.

Appendix Table A1. Bounding exercise: Effects of childcare quality on birthrates

	(1)	(2)	(3)
Log(Childcare Price)	-0.02 (0.02)	-0.01 (0.01)	-0.01 (0.02)
NAEYC Share	0.23 (0.45)	0.21 (0.40)	0.34 (0.33)
Mean (Price)	148.98	148.98	148.24
Mean (NAEYC Share, %)	6.81	6.81	7.12
N	2,678	2,678	1,948
Controls	No	Yes	Yes

* p < 0.10, ** p < 0.05, *** p < 0.01

Notes: Data: NVSS birth records, SEER county population counts, NDCP childcare prices, National Association for the Education of Young Children (NAEYC), Census County Business Patterns, for 2017-2019, 2021-2022. This table shows estimated effects of childcare price and quality on the log of the birthrate per 1000 women aged 20 to 44 from OLS regressions. Columns (1) and (2) include all states in the sample. Column (3) is sensitivity analysis that drops AR, AZ, CA, IL, MD, ME, SC, and TN. “Log(Childcare Price)” is the log of the median weekly price for full-time care at a childcare centre averaged for 0 to 2 year olds, winsorised at the 99th percentile. “NAEYC Share” is the number of NAEYC accredited childcare programmes out of the number of childcare establishments in the state. Columns (2) and (3) control for county median earnings, the unemployment rate, female labour force participation, the male-female ratio, a housing price index, and racial and ethnic composition. All models control for county and year fixed effects. Estimates weighted by births to women aged 20 to 44. Standard errors, clustered at the state-year level, in parentheses.

Appendix Table A2. Bounding exercise: Effects of childcare regulations on quality

	(1)	(2)
Maximum Group Size	-0.10*** (0.01)	-0.10*** (0.02)
Mean (Maximum Group Size)	16.74	16.26
Mean (NAEYC Share, %)	6.85	6.88
N	7,175	5,305

* p < 0.10, ** p < 0.05, *** p < 0.01

Notes: Data: Hand-collected maximum group size data, National Association for the Education of Young Children (NAEYC), Census County Business Patterns, for 2017-2019, 2021-2022. This table shows estimated effects of childcare regulations on childcare quality from OLS regressions. Coefficients and standard errors multiplied by 100. Column (1) includes all states in the sample. Column (2) is sensitivity analysis that drops AR, AZ, CA, IL, MD, ME, SC, and TN. “Maximum Group Size” is the state-level maximum group size average for 0 to 2 year olds. “NAEYC Share” is the number of NAEYC accredited childcare programmes out of the number of childcare establishments in the state. Both models control for county median earnings, the unemployment rate, female labour force participation, the male-female ratio, a housing price index, and racial and ethnic composition. Both models control for county and year fixed effects. Standard errors, clustered at the state-year level, in parentheses.

I now have estimates for the effect of childcare quality on birthrates ($\hat{\delta}$), the effect of the instrument (childcare regulations) on childcare quality ($\hat{\omega}$), and the effect of the instrument

on childcare prices ($\hat{\mu}$) (recall that we already have the estimator $\hat{\mu}$ in hand, as this is the first stage coefficient shown in Table 2.) Thus I can estimate the size of the potential bias, as shown below. Recall from Table 4 that my estimate for $\hat{\beta}_{IV}$ is -0.57. Both approaches deliver bias terms of similar magnitudes.

Maximum group size instrument:

$$\hat{\beta}_{IV} - 0.215 \cdot \frac{-0.000951}{-0.01} \longrightarrow \beta$$

$$\Rightarrow \beta \in [\hat{\beta} - 0.0316, \hat{\beta}]$$

Staff-child ratio instrument:

$$\hat{\beta}_{IV} - 0.215 \cdot \frac{0.639}{3.99} \longrightarrow \beta$$

$$\Rightarrow \beta \in [\hat{\beta} - 0.0344, \hat{\beta}]$$

B Appendix B: Theoretical model

B.1 Optimisation problem

The woman's optimisation problem is:

$$\max_{\{c,n,s\}} u = c_1 + c_2 + (2n_1 + n_2)v - \mu((1 - s_1)^\gamma n_1 + (1 - s_2)^\gamma n_2) \quad (9)$$

She is subject to the following budget and time constraints:

$$c_1 + ps_1n_1 \leq w(1 - (1 - s_1)n_1) \quad (10)$$

$$c_2 + ps_2n_2 \leq w(1 + \alpha h_1)(1 - (1 - s_2)n_2) \quad (11)$$

Where h_1 is the woman's work hours in $t = 1$: $h_1 = 1 - (1 - s_1)n_1$.

$$1 - (1 - s_1)n_1 \geq 0 \quad (12)$$

$$1 - (1 - s_2)n_2 \geq 0 \quad (13)$$

Equations 10 and 11 are the budget constraints for periods one and two respectively. The budget constraints reflect that the woman can spend on consumption and private childcare, and that this expenditure must be at most her labour income in that period. Equations 12 and 13 are the mother's time constraints for each period. Her time spent on employment and self-childcare must be less than or equal to one unit of time.

B.2 Optimal choice

B.2.1 Optimal share of childcare

We can derive the optimal share of private childcare purchased in each period (s_1, s_2) . Conditional on having a child in the relevant period, the first order conditions for s_1, s_2 are given by:

$$(w - p) + \mu\gamma(1 - s_1)^{\gamma-1} = 0$$

$$(w(1 + \alpha h_1) - p) + \mu\gamma(1 - s_2)^{\gamma-1} = 0$$

We can then derive the optimal share of childcare in periods one and two:

$$s_1^* = \begin{cases} 0, & \text{if } p > \mu\gamma + w, \\ 1 - \left(\frac{(p-w)}{\mu\gamma} \right)^{\frac{1}{\gamma-1}}, & \text{if } w \leq p \leq \mu\gamma + w, \\ 1, & \text{if } p < w. \end{cases}$$

$$s_2^* = \begin{cases} 0, & \text{if } p > \mu\gamma + w(1 + \alpha h_1^*), \\ 1 - \left(\frac{(p-w(1+\alpha h_1^*))}{\mu\gamma} \right)^{\frac{1}{\gamma-1}}, & \text{if } w(1 + \alpha h_1^*) \leq p \leq \mu\gamma + w(1 + \alpha h_1^*), \\ 1, & \text{if } p < w(1 + \alpha h_1^*). \end{cases}$$

$$\text{and where } h_1^* = \begin{cases} 0, & \text{if } s_1^* = 0 \text{ and } n_1 = 1 \\ 1 - \left(\frac{(p-w)}{\mu\gamma} \right)^{\frac{1}{\gamma-1}}, & \text{if } s_1^* = 1 - \left(\frac{(p-w)}{\mu\gamma} \right)^{\frac{1}{\gamma-1}} \text{ and } n_1 = 1, \\ 1, & \text{if } s_1^* = 1 \text{ and } n_1 = 1, \text{ or } n_1 = 0 \end{cases}$$

Optimal s_1^*, s_2^* depend on the wage w , price of childcare p , productivity α , and dis-utility from self-childcare parameters γ and μ . An increase in the price of childcare reduces the share of private childcare purchased, while wage increases raise it. If the wage that period exceeds the childcare price, the woman will fully outsource childcare. Conversely, if prices are too high relative to wages, some women exit the labour market to provide full-time care for their child. Human capital accumulation interacts with these decisions: outsourcing childcare in period one allows for more labour supply, raising human capital and wages in period two, making outsourcing in the second period more attractive.

Note that if $p < w + \mu\gamma$, then $s_1^* > 0$. Since the period two wage $w_2 \equiv w(1 + \alpha h_1^*) \geq w$, it follows that $p < w_2 + \mu\gamma$ and thus $s_2^* > 0$. Moreover, holding p (and μ, γ) fixed, $s^*(\cdot)$ is weakly increasing in the effective wage, so $s_2^* \geq s_1^*$. Mothers with second period births will purchase a weakly larger share of private childcare than mothers who have a first period birth. Intuitively, this arises because the period 2 wage uplift makes outsourcing relatively more attractive.

B.2.2 Optimal fertility choices

As n takes discrete values of 0 or 1, we can set out equations for the woman's utility under each of the four fertility scenarios with s_1^* and s_2^* as above. Utility $u(n_1, n_2)$ for $n_1, n_2 \in \{0, 1\}$ is denoted by:

$$\begin{aligned} u(0, 0) &= w(2 + \alpha) \\ u(1, 0) &= w(1 - (1 - s_1^*)) - ps_1^* - \mu(1 - s_1^*)^\gamma + w(1 + \alpha) + 2v \\ u(0, 1) &= w + w(1 + \alpha)(1 - (1 - s_2^*)) - ps_2^* - \mu(1 - s_2^*)^\gamma + v \end{aligned}$$

$$\begin{aligned} u(1, 1) &= w(1 - (1 - s_1^*)) - ps_1^* + w(1 + \alpha h_1^*)(1 - (1 - s_2^*)) \\ &\quad - ps_2^* - \mu((1 - s_1^*)^\gamma + (1 - s_2^*)^\gamma) + 3v \end{aligned}$$

Based on optimal choices s_1^*, s_2^* (which depend on the wage w , price of childcare p , and disutility from self-childcare μ, γ), a woman will compare her utility in each fertility scenario to determine whether to have children, and how many to have.

B.2.3 The utility gain from having a child

We can denote the utility gain from having a child as:

$$\Delta u(1, 0) = u(1, 0) - u(0, 0) = 2v - w(1 - s_1^*) - ps_1^* - \mu(1 - s_1^*)^\gamma$$

$$\Delta u(0, 1) = u(0, 1) - u(0, 0) = v - w(1 + \alpha)(1 - s_2^*) - ps_2^* - \mu(1 - s_2^*)^\gamma$$

$$\Delta u(1, 1) = u(1, 1) - u(1, 0) = v - w(1 + \alpha h_1^*)(1 - s_2^*) - ps_2^* - \mu(1 - s_2^*)^\gamma + w\alpha(h_1^* - 1)$$

For women who do some self-childcare ($0 < s_i^* < 1$), they will have a child if the value, or

joy, from having a child, is greater than three components: (i) the opportunity cost of time, $w_t(1 - s_i^*)$, (ii) the cost of private childcare, ps_i^* , and (iii) the disutility of self-care, $\mu(1 - s_i^*)^\gamma$. Here $w_1 = w$ in $t = 1$ and $w_2 = w(1 + \alpha h_1^*)$ in $t = 2$. For women that purchase part-time childcare, the foregone wage is scaled by $(1 - s_i^*)$ and the childcare price by the share of private childcare s_i^* . As discussed above, increases in the wage that period and productivity raise the share of private childcare, so will increase the private childcare cost term but reduce the foregone wage term. Women who fully outsource ($s_i^* = 1$) only weigh the child value against the cost of private childcare. Each of the cases are discussed in more detail below.

Case 1: Stay at home mothers ($s_1^*, s_2^* = 0$). Mothers who will not outsource any childcare will have a child if the benefit from having a child outweighs the opportunity cost of foregone wages from full-time self-childcare. They will have a first and second period birth respectively when:

$$2v \geq w + \mu$$

$$v \geq w(1 + \alpha h_1^*) + \mu$$

Case 2: Mothers with part-time private childcare in both periods. These mothers will have a child if the benefit is greater than the foregone wages (weighted by share of self-childcare), cost of private childcare, and dis-utility from self-childcare. They will have a first and second period birth respectively when:

$$2v \geq w(1 - s_1^i) + ps_1^i + \mu(1 - s_1^i)^\gamma$$

$$v \geq w(1 + \alpha h_1^*)(1 - s_2^i) + ps_2^i + \mu(1 - s_2^i)^\gamma$$

where

$$s_1^i = 1 - \left(\frac{(p - w)}{\mu\gamma} \right)^{\frac{1}{\gamma-1}}, s_2^i = 1 - \left(\frac{(p - w(1 + \alpha h_1^*))}{\mu\gamma} \right)^{\frac{1}{\gamma-1}}$$

Case 3: Mothers with part-time private childcare in $t = 1$, full-time childcare in the second period in $t = 2$. These mothers will have a child in the first period if the benefit is greater than the foregone wages (weighted by share of self-childcare), cost of private childcare, and dis-utility from self-childcare. They will have a child in the second period if the benefit outweighs the cost of private childcare for that second period. They will have a first and second period birth respectively when:

$$2v \geq w(1 - s_1^i) + ps_1^i + \mu(1 - s_1^i)^\gamma$$

$$v \geq p$$

where

$$s_1^i = 1 - \left(\frac{(p - w)}{\mu\gamma} \right)^{\frac{1}{\gamma-1}}$$

Case 4: Mothers who fully outsource childcare ($s_1^, s_2^* = 1$).* Mothers who purchase as much as private childcare as possible have a child in the first and second period if the benefit is greater than the monetary cost of private childcare. They will have a first and second period birth respectively when:

$$2v \geq p$$

$$v \geq p$$

B.3 Response to price changes

B.3.1 Utility responses

Next, I consider how utility at different fertility outcomes varies with respect to the childcare price:

$$\frac{\partial u(1,0)}{\partial p} = -s_1^* = \begin{cases} 0, & \text{if } p \geq \mu\gamma + w, \\ -\left(1 - \left(\frac{p-w}{\mu\gamma}\right)^{\frac{1}{\gamma-1}}\right), & \text{if } w \leq p \leq \mu\gamma + w, \\ -1, & \text{if } p \leq w. \end{cases}$$

$$\frac{\partial u(0,1)}{\partial p} = -s_2^* = \begin{cases} 0, & \text{if } p \geq \mu\gamma + w(1+\alpha), \\ -\left(1 - \left(\frac{p-w(1+\alpha)}{\mu\gamma}\right)^{\frac{1}{\gamma-1}}\right), & \text{if } w(1+\alpha) \leq p \leq \mu\gamma + w(1+\alpha), \\ -1, & \text{if } p \leq w(1+\alpha). \end{cases}$$

$$\begin{aligned} \frac{\partial u(1,1)}{\partial p} = -(s_1^* + s_2^*) = & - \left(\begin{cases} 0, & \text{if } p \geq w + \mu\gamma, \\ 1 - \left(\frac{p-w}{\mu\gamma}\right)^{\frac{1}{\gamma-1}}, & \text{if } w \leq p \leq w + \mu\gamma, \\ 1, & \text{if } p \leq w \end{cases} \right. \\ & \left. + \begin{pmatrix} 0, & \text{if } p \geq w(1+\alpha h_1^*) + m\mu\gamma, \\ 1 - \left(\frac{p-w(1+\alpha h_1^*)}{\mu\gamma}\right)^{\frac{1}{\gamma-1}}, & \text{if } w(1+\alpha h_1^*) \leq p \leq w(1+\alpha h_1^*) + \mu\gamma, \\ 1, & \text{if } p \leq w(1+\alpha h_1^*) \end{pmatrix} \right) \end{aligned}$$

Number of children By comparing $\frac{\partial u(1,0)}{\partial p}$ and $\frac{\partial u(0,1)}{\partial p}$, and recalling that $s_2^* \geq s_1^*$, we see that second period births (to older women) are more price responsive than first period births (to younger women) for interior solutions. The wage uplift in period 2 raises s_2^* , which in turn increases the price sensitivity of utility. Delaying birth therefore makes utility more price sensitive: human capital accumulation raises wages, leading mothers to outsource more childcare in the second period. This higher level of outsourcing exposes mothers more to price changes.

Birth timing amongst one-child mothers A comparison of $\frac{\partial u(1,0)}{\partial p}$ and $\frac{\partial u(1,1)}{\partial p}$ yields a straightforward prediction. Consider mothers who purchase some share of private childcare in the first period. As shown above, they will purchase at least that amount in the second period. Thus mothers who have two children, rather than one, will have utility that is more price sensitive because they face childcare costs in both periods.

B.3.2 Child-state responses

Sufficiently large price changes can also shift mothers between the optimal s_1^* and s_2^* values. For example, a price increase can move a mother towards part-time private childcare or full-time self-childcare. Ultimately, we are interested in how the probability of choosing one of the four utility states $(u(0,0), u(1,0), u(0,1), u(1,1))$ varies with respect to the price of childcare, p . To capture reallocation across fertility states, I model the probability of being in one of the four states using multinomial logit. This requires the assumption that the unobserved taste errors for each state follow independent extreme-value distributions. This delivers tractable choice probabilities, but is a strong assumption as it rules out correlation in unobserved preferences across states.

$$P_{n_1, n_2} = \frac{e^{u_{n_1, n_2}}}{\sum_{k \in \{(0,0), (1,0), (0,1), (1,1)\}} e^{u_k}}$$

where each utility u_{n_1, n_2} state is evaluated at the optimal childcare shares s_1^*, s_2^* .

Using the fact that:

$$\frac{\partial P_{n_1, n_2}}{\partial u_{k_1, k_2}} = P_{n_1, n_2} (\mathbf{1}\{(n_1, n_2) = (k_1, k_2)\} - P_{k_1, k_2}),$$

We can derive the following:

$$\frac{dP_{n_1, n_2}}{dp} = \sum_{k \in \{(0,0), (1,0), (0,1), (1,1)\}} \frac{\partial P_{n_1, n_2}}{\partial u_k} \frac{du_k}{dp} = P_{n_1, n_2} \left(\frac{du_{n_1, n_2}}{dp} - \sum_{k \in \{(0,0), (1,0), (0,1), (1,1)\}} P_k \frac{du_k}{dp} \right)$$

Recall that the within state derivatives are given by:

$$\frac{du_{0,0}}{dp} = 0, \quad \frac{du_{1,0}}{dp} = -s_1^*, \quad \frac{du_{0,1}}{dp} = -s_2^*, \quad \frac{du_{1,1}}{dp} = -(s_1^* + s_2^*).$$

Which we can use to derive:

$$\begin{aligned} \frac{dP_{00}}{dp} &= P_{00} \left[s_1^* (P_{10} + P_{11}) + s_2^* (P_{01} + P_{11}) \right], \\ \frac{dP_{10}}{dp} &= P_{10} \left[s_1^* (P_{10} + P_{11} - 1) + s_2^* (P_{01} + P_{11}) \right], \\ \frac{dP_{01}}{dp} &= P_{01} \left[s_1^* (P_{10} + P_{11}) + s_2^* (P_{01} + P_{11} - 1) \right], \\ \frac{dP_{11}}{dp} &= P_{11} \left[s_1^* (P_{10} + P_{11} - 1) + s_2^* (P_{01} + P_{11} - 1) \right]. \end{aligned}$$

With the optimal private childcare shares s_1^*, s_2^* as before.

Having no children First, observe that $\frac{dP_{00}}{dp} \geq 0$ whenever $s_1^*(p) > 0$ or $s_2^*(p) > 0$. A childcare price increase weakly raises the probability of having no children.

Number of children Computing the price elasticities $\eta_{\{n_1, n_2\}, p}$ aids a comparison of the one-child versus two-child states:

$$\begin{aligned} \eta_{10, p} &= p \left[s_1^* (P_{10} + P_{11} - 1) + s_2^* (P_{01} + P_{11}) \right], \\ \eta_{11, p} &= p \left[s_1^* (P_{10} + P_{11} - 1) + s_2^* (P_{01} + P_{11} - 1) \right] \end{aligned}$$

$$\begin{aligned}
\eta_{11,p} - \eta_{10,p} &= p \left\{ s_1^* [(P_{10} + P_{11} - 1) - (P_{10} + P_{11} - 1)] \right. \\
&\quad \left. + s_2^* [(P_{01} + P_{11} - 1) - (P_{01} + P_{11})] \right\} \\
&= -p s_2^*
\end{aligned}$$

The probability of having two children has a (weakly) more negative price elasticity than that of having one child, because of the additional private care exposure.

Birth timing amongst one-child mothers To evaluate how price changes impact delays to birth, I consider birth timing amongst mothers with one child. Conditional on having one child, the probability of choosing a first-period birth is: $P_{10|1} \equiv \frac{P_{10}}{P_{10} + P_{01}}$. A marginal increase in the childcare price p shifts this probability by:

$$\frac{dP_{10|1}}{dp} = \frac{P_{01} \frac{dP_{10}}{dp} - P_{10} \frac{dP_{01}}{dp}}{(P_{10} + P_{01})^2} = - \frac{P_{10}P_{01}}{(P_{10} + P_{01})^2} (s_1^* - s_2^*) = - P_{10|1}(1 - P_{10|1})(s_1^* - s_2^*)$$

By the result established above that $s_2^*(p) \geq s_1^*(p)$, it follows that

$$\frac{dP_{10|1}}{dp} \geq 0 \quad \text{and} \quad \frac{dP_{01|1}}{dp} = -\frac{dP_{10|1}}{dp} \leq 0$$

A marginal price increase reduces the probability of a second-period birth; it shifts timing toward the first period amongst one-child mothers. The term $P_{10|1}(1 - P_{10|1})$ scales the magnitude of the derivative; it captures how indifferent between the two timings the mother is, and is small when one option dominates. If $s_1^* = s_2^*$, birth timing is insensitive to price.

B.4 Extension: taste for self-caring for the child

The model presented above incorporates a distaste for self-childcare, particularly in large quantities. The dis-utility term captures that a share of mothers have a preference for a career, and that caring for their child at home limits their ability to progress in the workplace. However, not all parents may feel this way about caring for their own child, and in fact may

enjoy spending a portion of their working hours caring for their child. The model can easily be adjusted to reflect a taste for self-childcare. We simply flip the sign of the dis-utility parameter and require that $0 < \gamma < 1$ so that the taste for self-childcare is concave: the marginal benefit from self-childcare is diminishing. This adjustment does not change the model predictions, but pushes the interior solution (part-time childcare) away from more private childcare and towards more self child-care.

We can replace Equation 6.1 with the following utility function:

$$u = c_1 + c_2 + (2n_1 + n_2)v + \mu((1 - s_1)^\gamma n_1 + (1 - s_2)^\gamma n_2)$$

The budget and time constraints remain the same. To derive the optimal share of private childcare purchased in each period (s_1, s_2) we take first order conditions:

$$(w - p) - \mu\gamma(1 - s_1)^{\gamma-1} = 0$$

$$(w(1 + \alpha h_1^*) - p) - \mu\gamma(1 - s_2)^{\gamma-1} = 0$$

From these we determine the optimal share of childcare in periods one and two:

$$s_1^* = \begin{cases} 0, & \text{if } p \geq w - \mu\gamma, \\ 1 - \left(\frac{(w - p)}{\mu\gamma} \right)^{\frac{1}{\gamma-1}}, & \text{if } p < w - \mu\gamma. \end{cases}$$

$$s_2^* = \begin{cases} 0, & \text{if } p \geq w(1 + \alpha h_1^*) - \mu\gamma, \\ 1 - \left(\frac{(w(1 + \alpha h_1^*) - p)}{\mu\gamma} \right)^{\frac{1}{\gamma-1}}, & \text{if } p < w(1 + \alpha h_1^*) - \mu\gamma. \end{cases}$$

$$\text{and where } h_1^* = \begin{cases} 1, & \text{if } s_1^* = 0 \text{ and } n_1 = 1 \\ 1 - \left(\frac{(w-p)}{\mu\gamma}\right)^{\frac{1}{\gamma-1}}, & \text{if } s_1^* = 1 - \left(\frac{(w-p)}{\mu\gamma}\right)^{\frac{1}{\gamma-1}} \text{ and } n_1 = 1, \\ 1, & \text{if } n_1 = 0 \end{cases}$$

Note that due to the positive utility, or joy, that the mother receives from self-childcare, mothers never choose full outsourcing ($s_i^* = 1$). Instead they either select part-time private childcare or provide all care themselves.

As before, increases in the price of childcare reduce the share of private childcare purchased. Increases in the wage and productivity lead the share of private childcare to rise. Because the wage in period two is at least as high as in period one ($w(1 + \alpha h_1^*) \geq w$), if a mother chooses part-time private childcare in the first period she will also choose part-time in the second.

Examining how mothers now respond to price changes, we see that:

$$\begin{aligned} \frac{\partial u(1,0)}{\partial p} = -s_1^* &= \begin{cases} 0, & \text{if } p \geq w - \mu\gamma, \\ -\left(1 - \frac{(w-p)}{\mu\gamma}\right)^{\frac{1}{\gamma-1}}, & \text{if } p < w - \mu\gamma. \end{cases} \\ \frac{\partial u(0,1)}{\partial p} = -s_2^* &= \begin{cases} 0, & \text{if } p \geq w(1 + \alpha) - \mu\gamma, \\ -\left(1 - \frac{(w(1 + \alpha) - p)}{\mu\gamma}\right)^{\frac{1}{\gamma-1}}, & \text{if } p < w(1 + \alpha) - \mu\gamma. \end{cases} \\ \frac{\partial u(1,1)}{\partial p} = -(s_1^* + s_2^*) &= -\left(\begin{cases} 0, & \text{if } p \geq w - \mu\gamma, \\ 1 - \left(\frac{(w-p)}{\mu\gamma}\right)^{\frac{1}{\gamma-1}}, & \text{if } p < w - \mu\gamma \end{cases} \right. \\ &\quad \left. + \begin{cases} 0, & \text{if } p \geq w(1 + \alpha h_1^*) - \mu\gamma, \\ 1 - \left(\frac{(w(1 + \alpha h_1^*) - p)}{\mu\gamma}\right)^{\frac{1}{\gamma-1}}, & \text{if } p < w(1 + \alpha h_1^*) - \mu\gamma \end{cases} \right) \end{aligned}$$

The model predictions remain unchanged with this modification to the utility function. Second births (to older women) are more price responsive than first period births (to younger women). Again, this reflects the higher effective wage in period two, which leads to more outsourcing of childcare. This greater reliance on private childcare increases the magnitude of price sensitivity. Similarly, women with higher wages are more responsive to childcare price changes, as they outsource more childcare. Finally, for mothers with two children, utility is more price sensitive than for those with one child, since private childcare is purchased in both periods.

C Appendix C: Informal childcare analysis

In Section 5.3 I discuss interpretation of the childcare price variable and show that childcare centre prices are a proxy for close childcare substitutes by evidencing that stricter childcare regulations also increase prices of informal childcare. In this analysis, to measure prices of informal childcare, I use wages of nannies reported in the CPS for the years 2009 to 2022. I identify nannies by restricting to employed individuals with 2010 Census Occupational Classification code “4600” for childcare workers, and industry codes “9290” (private households) or “7580” (employment services). The outcome is weekly earnings. Earnings data are collected from a quarter of the total CPS sample, and represent earnings before taxes and another deductions.

I use a Difference-in-Differences (DiD) strategy, exploiting variation in regulatory changes across time and states. I evaluate the effect of changes in the continuous value of the maximum group size or staff-child ratio in Delaware, Vermont, Nevada, Louisiana, Utah, Virginia, Arizona, Idaho, and South Carolina between 2009 and 2022.¹⁸ In my sample, the change is either an increase/decrease in the staff-child ratio, an increase/decrease in the maximum group size, or an introduction of a maximum group size. I control for state¹⁹ and year fixed effects. In an alternative specification I include month fixed effects to control for seasonal patterns; results are consistent. I also control for the individual’s age, age squared, sex, race and ethnicity, immigrant status, education, whether they live in a metropolitan area, and part-time status.

¹⁸The main analysis limits my sample of states with regulatory changes due to missing-ness in the childcare centre price data, but in this CPS analysis I am able to take advantage of additional regulatory changes.

¹⁹State is the lowest geographical unit available in these data.

D Appendix Tables

Appendix Table D1. Robustness of estimates to clustering standard errors at state level

	20-44	20-24	25-29	30-34	35-39	40-44
<i>IV</i>						
Log (Childcare Price)	-0.57*** (0.21)	-0.14 (0.21)	-0.12 (0.23)	-0.73*** (0.23)	-0.87*** (0.30)	-0.80*** (0.31)
<i>Reduced Form</i>						
Maximum Group Size	0.37*** (0.11)	0.10 (0.15)	0.08 (0.16)	0.46*** (0.11)	0.54*** (0.13)	0.49*** (0.11)
Mean	69.76	97.57	122.42	89.99	37.71	7.45

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: $N = 7,495$. Data: NVSS birth records, SEER county population counts, NDCP childcare prices, hand-collected maximum group size data, for 2010-2019, 2021-2022. This table shows the estimated effects of log childcare prices on the log of birthrates per 1000 women. Reduced form coefficients and standard errors multiplied by 100. “20-44” is the log of the county-level birthrate per 1000 women aged 20 to 44, “20-24” the birthrate per 1000 women aged 20-24, and so on. “Childcare price” is the median weekly price for full-time care at a childcare centre averaged for 0 to 2 year olds, winsorised at the 99th percentile. “Maximum Group Size” is the state-level maximum group size average for 0 to 2 year olds. All models control for county median earnings, the unemployment rate, female labour force participation, the male-female ratio, a housing price index, racial and ethnic composition, and county and year fixed effects. Estimates weighted by births to that age band. Standard errors, clustered at the state level, in parentheses.

Appendix Table D2. First stage

	Robustness sample (NV, VT, DE)		Main sample (NV, VT)	
	Price	Log(Price) $\times 100$	Price	Log(Price) $\times 100$
Maximum Group Size	-0.04 (0.16)	-0.08 (0.10)	32.50*** (3.29)	15.00*** (1.92)
Staff-Child Ratio	735.31*** (3.40)	398.62*** (34.88)	24 921.58*** (2401.51)	11 616.65*** (1440.98)
Mean (Childcare Price)	146.45	146.45	146.57	146.57
Mean (Maximum Group Size)	16.26	16.26	16.10	16.10
Mean (Staff-Child Ratio)	0.20	0.20	0.20	0.20
F-stat	24 001.13	92.33	59.72	37.70
N	8,094	8,094	7,495	7,495

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Data: NDCP childcare prices, hand-collected regulation data, for 2009-2019, 2021-2022 (Robustness sample, with regulatory changes in NV, VT, and DE), 2010-2019, 2021-2022 (Main sample, with regulatory changes in NV and VT). This table shows the first stage of childcare prices on the maximum group size and staff-child ratio. “Childcare price” is the median weekly price for full-time care at a childcare centre averaged for 0 to 2 year olds, adjusted to 2010 dollars and winsorised at the 99th percentile. The Log(Price) is unadjusted. “Maximum Group Size” is the state-level maximum group size average for 0 to 2 year olds. “Staff-Child Ratio” is the state-level staff-child ratio average for 0 to 2 year olds. “F-stat” is the first-stage Kleibergen-Paap Wald F statistic. All models control for county median earnings, the unemployment rate, female labour force participation, the male-female ratio, a housing price index, racial and ethnic composition, and county and year fixed effects. Estimates weighted by births to women aged 20-44. Standard errors, clustered at the state-year level, in parentheses.

Appendix Table D3. Robustness of estimates to using interacted instrument

	20-44	20-24	25-29	30-34	35-39	40-44
<i>IV</i>						
Log(Childcare Price)	-0.55*** (0.14)	-0.17 (0.17)	-0.06 (0.20)	-0.74*** (0.15)	-0.81*** (0.21)	-0.90*** (0.20)
<i>Reduced Form</i>						
Below-Median Income $\times$ Maximum Group Size	0.37*** (0.13)	0.09 (0.17)	0.09 (0.13)	0.46*** (0.12)	0.54*** (0.12)	0.47*** (0.08)
Mean	69.76	97.57	122.42	89.99	37.71	7.45

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: N = 7,495. Data: NVSS birth records, SEER county population counts, NDCP childcare prices, hand-collected maximum group size data, for 2010-2019, 2021-2022. This table shows the estimated effects of log childcare prices on the log of birthrates per 1000 women, using a below-median-income interacted instrument. The instrument interacts maximum group size with an indicator for counties below the 2009 median household income. Reduced form coefficients and standard errors multiplied by 100. “20-44” is the log of the county-level birthrate per 1000 women aged 20 to 44, “20-24” the log of the birthrate per 1000 women aged 20-24, and so on. “Childcare price” is the median weekly price for full-time care at a childcare centre averaged for 0 to 2 year olds, winsorised at the 99th percentile. “Maximum Group Size” is the state-level maximum group size average for 0 to 2 year olds. All models control for county median earnings, the unemployment rate, female labour force participation, the male-female ratio, a housing price index, racial and ethnic composition, and county and year fixed effects. Estimates weighted by births to that age band. Standard errors, clustered at the state-year level, in parentheses.

Appendix Table D4. Robustness of IV estimates to using multiple instruments

	20-44	20-24	25-29	30-34	35-39	40-44
<i>Main sample (NV, VT)</i>						
Log (Childcare Price)	-0.57*** (0.10)	-0.57 (0.49)	-0.24 (0.15)	-0.62*** (0.15)	-0.72*** (0.21)	-0.51 (0.31)
<i>Robustness sample (NV, VT, DE)</i>						
Log (Childcare Price)	-0.64*** (0.16)	-0.28** (0.12)	-0.22 (0.18)	-0.80*** (0.17)	-0.93*** (0.21)	-0.70** (0.24)
Mean	69.78	99.21	122.42	89.62	37.34	7.40

* p < 0.10, ** p < 0.05, *** p < 0.01.

Notes: N = 7,495 (Main sample, 2010-2019, 2021-2022 with regulatory changes in NV and VT), 8,094 (Robustness sample, 2009-2019, 2021-2022 with regulatory changes in NV, VT, and DE). Data: NVSS birth records, SEER county population counts, NDCP childcare prices, hand-collected maximum group size data, for 2009-2019, 2021-2022. This table shows the estimated effects of log childcare prices on the log of birthrates per 1000 women. Coefficients and standard errors multiplied by 100. “20-44” is the log of the county-level birthrate per 1000 women aged 20 to 44, “20-24” the log of the birthrate per 1000 women aged 20-24, and so on. “Log (Childcare Price)” is the log of the median weekly price for full-time care at a childcare centre averaged for 0 to 2 year olds, winsorised at the 99th percentile. Two instruments are used: “Maximum Group Size”, the state-level maximum group size average for 0 to 2 year olds, and “Child-Staff Ratio”, the state-level child-staff ratio average for 0 to 2 year olds. All models control for county median earnings, the unemployment rate, female labour force participation, the male-female ratio, a housing price index, racial and ethnic composition, and county and year fixed effects. Estimates weighted by births to that age band. Standard errors, clustered at the state-year level, in parentheses.

Appendix Table D5. Effects of childcare regulations on nannies’ weekly wages

	Weekly Earnings		Log(Earnings) × 100	
Maximum Group Size	-10.90** (3.93)	-11.44** (4.08)	-0.90* (0.44)	-1.14* (0.53)
Staff-Child Ratio	-2102.19 (1382.43)	-1636.03 (1511.40)	-440.36 (654.32)	-193.97 (765.47)
State + Year FE	Yes	Yes	Yes	Yes
Month FE	No	Yes	No	Yes
Mean (Weekly Earnings)	370.22	370.22	370.22	370.22
Mean (Maximum Group Size)	18.01	18.01	18.01	18.01
Mean (Staff-Child Ratio)	0.22	0.22	0.22	0.22

* p < 0.10, ** p < 0.05, *** p < 0.01.

Notes: N = 2,238. Data: CPS earnings, hand-collected regulation data, for 2009-2022. This table shows the effect of childcare regulations on weekly earnings of nannies, with separate regressions for the two regulations. “Weekly Earnings” is weekly earnings in the prior week for wage and salary workers whose occupation is childcare (4600), and whose industry is private household (9290) or employment services (7580). “Maximum Group Size” is the state-level maximum group size average for 0 to 2 year olds. “Staff-Child Ratio” is the state-level child-staff ratio average for 0 to 2 year olds. All models include state fixed effects and controls for age, age squared, sex, race, ethnicity, immigrant status, education, metropolitan status, and part-time status. Estimates weighted by CPS earnings weights. Standard errors, clustered at the state and year level, in parentheses.

Appendix Table D6. Within-MRR heterogeneity

	Mean	SD	CV
Log of Earnings	10.29	0.09	0.01
Log of Median Household Income	10.86	0.10	0.01
Unemployment Rate	6.40	1.12	0.18
Male-Female Ratio	1.06	0.08	0.07
Male LFP (%)	80.29	4.71	0.06
Female LFP (%)	71.42	2.66	0.04
White (%)	81.29	5.74	0.08
Black (%)	9.57	4.00	0.46
Asian (%)	2.12	0.87	0.44
Hispanic (%)	14.42	4.02	0.37

Notes: Data: NDCP for 2010-2019, 2021-2022. Market rate areas (MRRs) are identified by counties sharing an identical reported childcare price within a state-year. For each row, the mean, standard deviation, and coefficient of variation (SD / mean) are computed across the MRR's member counties, weighted by each county's total population. Statistics for all MRRs with at least 2 counties (1123 of 1897 MRRs are singleton and excluded).

Appendix Table D7. Robustness of estimates to controlling for share of childcare establishments

	20-44	20-24	25-29	30-34	35-39	40-44
<i>OLS</i>						
Log(Childcare Price)	0.01 (0.02)	0.01 (0.04)	0.03 (0.03)	0.01 (0.02)	0.01 (0.02)	-0.02 (0.03)
<i>IV</i>						
Log(Childcare Price)	-0.53*** (0.13)	-0.11 (0.17)	-0.10 (0.19)	-0.67*** (0.14)	-0.79*** (0.20)	-0.75*** (0.19)
<i>Reduced Form</i>						
Maximum Group Size	0.35*** (0.12)	0.07 (0.11)	0.06 (0.12)	0.43*** (0.12)	0.50*** (0.10)	0.47*** (0.08)
Mean	69.76	97.57	122.42	89.99	37.71	7.45

* p < 0.10, ** p < 0.05, *** p < 0.01.

Notes: N = 7,495. Data: NVSS birth records, SEER county population counts, NDCP childcare prices, hand-collected maximum group size data, for 2010-2019, 2021-2022. This table shows the estimated effects of log childcare prices on the log of birthrates per 1000 women. Reduced form coefficients and standard errors multiplied by 100. "20-44" is the log of the county-level birthrate per 1000 women aged 20 to 44, "20-24" the log of the birthrate per 1000 women aged 20-24, and so on. "Childcare price" is the median weekly price for full-time care at a childcare centre averaged for 0 to 2 year olds, winsorised at the 99th percentile. "Maximum Group Size" is the state-level maximum group size average for 0 to 2 year olds. All models control for county median earnings, the unemployment rate, female labour force participation, the male-female ratio, a housing price index, racial and ethnic composition, the share of childcare establishments, and county and year fixed effects. Estimates weighted by births to that age band. Standard errors, clustered at the state-year level, in parentheses.

Appendix Table D8. Robustness of estimates to controlling for staff turnover

	20-44	20-24	25-29	30-34	35-39	40-44
<i>OLS</i>						
Log(Childcare Price)	0.01 (0.02)	0.03 (0.04)	0.03 (0.03)	0.02 (0.02)	0.02 (0.02)	-0.02 (0.03)
<i>IV</i>						
Log(Childcare Price)	-0.59*** (0.14)	-0.17 (0.18)	-0.23 (0.17)	-0.75*** (0.13)	-0.86*** (0.19)	-0.93*** (0.17)
<i>Reduced Form</i>						
Maximum Group Size	0.40*** (0.12)	0.12 (0.12)	0.16 (0.12)	0.49*** (0.12)	0.55*** (0.11)	0.60*** (0.06)
Mean	69.76	97.57	122.42	89.99	37.71	7.45

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: $N = 7,495$. Data: NVSS birth records, SEER county population counts, NDCP childcare prices, hand-collected maximum group size data, for 2010-2019, 2021-2022. This table shows the estimated effects of log childcare prices on the log of birthrates per 1000 women. Reduced form coefficients and standard errors multiplied by 100. “20-44” is the log of the county-level birthrate per 1000 women aged 20 to 44, “20-24” the log of the birthrate per 1000 women aged 20-24, and so on. “Childcare price” is the median weekly price for full-time care at a childcare centre averaged for 0 to 2 year olds, winsorised at the 99th percentile. “Maximum Group Size” is the state-level maximum group size average for 0 to 2 year olds. All models control for county median earnings, the unemployment rate, female labour force participation, the male-female ratio, a housing price index, racial and ethnic composition, the share of childcare establishments, and county and year fixed effects. Estimates weighted by births to that age band. Standard errors, clustered at the state-year level, in parentheses.

Appendix Table D9. Robustness of estimates to dropping bordering counties

	20-44	20-24	25-29	30-34	35-39	40-44
<i>OLS</i>						
Log (Childcare Price)	0.01 (0.02)	0.02 (0.04)	0.03 (0.03)	0.02 (0.02)	0.02 (0.02)	-0.01 (0.03)
<i>IV</i>						
Log (Childcare Price)	-0.57*** (0.14)	-0.15 (0.18)	-0.13 (0.20)	-0.73*** (0.15)	-0.87*** (0.21)	-0.79*** (0.20)
<i>Reduced Form</i>						
Maximum Group Size	0.37*** (0.13)	0.10 (0.12)	0.09 (0.13)	0.46*** (0.12)	0.54*** (0.11)	0.48*** (0.08)
Mean	69.70	97.53	122.30	89.94	37.62	7.41

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: $N = 7,395$. Data: NVSS birth records, SEER county population counts, NDCP childcare prices, hand-collected maximum group size data, for 2010-2019, 2021-2022. This table shows the estimated effects of log childcare prices on the log of birthrates per 1000 women. The sample does not contain counties bordering Nevada and Vermont. Reduced form coefficients and standard errors multiplied by 100. “20-44” is the log of the county-level birthrate per 1000 women aged 20 to 44, “20-24” the log of the birthrate per 1000 women aged 20-24, and so on. “Childcare price” is the median weekly price for full-time care at a childcare centre averaged for 0 to 2 year olds, winsorised at the 99th percentile. “Maximum Group Size” is the state-level maximum group size average for 0 to 2 year olds. All models control for county median earnings, the unemployment rate, female labour force participation, the male-female ratio, a housing price index, racial and ethnic composition, and county and year fixed effects. Estimates weighted by births to that age band. Standard errors, clustered at the state-year level, in parentheses.

Appendix Table D10. Robustness of estimates to weighting by population

	20-44	20-24	25-29	30-34	35-39	40-44
<i>OLS</i>						
Log (Childcare Price)	-0.00 (0.03)	0.00 (0.04)	0.03 (0.04)	-0.00 (0.03)	0.01 (0.03)	-0.03 (0.03)
<i>IV</i>						
Log (Childcare Price)	-0.63*** (0.15)	-0.30 (0.19)	-0.27 (0.20)	-0.81*** (0.15)	-0.88*** (0.21)	-0.79*** (0.21)
<i>Reduced Form</i>						
Maximum Group Size	0.38*** (0.11)	0.18* (0.11)	0.16 (0.12)	0.48*** (0.08)	0.54*** (0.11)	0.49*** (0.12)
Mean	69.76	97.57	122.42	89.99	37.71	7.45

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: $N = 7,495$. Data: NVSS birth records, SEER county population counts, NDCP childcare prices, hand-collected maximum group size data, for 2010-2019, 2021-2022. This table shows the estimated effects of log childcare prices on the log of birthrates per 1000 women. Reduced form coefficients and standard errors multiplied by 100. “20-44” is the log of the county-level birthrate per 1000 women aged 20 to 44, “20-24” the log of the birthrate per 1000 women aged 20-24, and so on. “Childcare price” is the median weekly price for full-time care at a childcare centre averaged for 0 to 2 year olds, winsorised at the 99th percentile. “Maximum Group Size” is the state-level maximum group size average for 0 to 2 year olds. All models control for county median earnings, the unemployment rate, female labour force participation, the male-female ratio, a housing price index, racial and ethnic composition, and county and year fixed effects. Estimates weighted by population in that age band. Standard errors, clustered at the state-year level, in parentheses.

Appendix Table D11. Robustness of IV estimates to leave-one-state-out analysis

Dropped State	Coefficient	Standard Error	P-value
<i>All states</i>	-0.57	(0.14)	0.00
AK	-0.57	(0.14)	0.00
AL	-0.57	(0.14)	0.00
AZ	-0.63	(0.15)	0.00
CA	-0.70	(0.09)	0.00
CT	-0.56	(0.14)	0.00
DE	-0.57	(0.14)	0.00
FL	-0.37	(0.20)	0.09
ID	-0.57	(0.14)	0.00
IL	-0.43	(0.13)	0.01
KS	-0.57	(0.14)	0.00
KY	-0.57	(0.14)	0.00
LA	-0.57	(0.14)	0.00
MA	-0.63	(0.13)	0.00
MD	-0.56	(0.14)	0.00
ME	-0.57	(0.14)	0.00
MN	-0.58	(0.15)	0.00
NE	-0.57	(0.15)	0.00
NV	-0.57	(0.15)	0.00
OH	-0.56	(0.13)	0.00
OR	-0.58	(0.13)	0.00
PA	-0.57	(0.14)	0.00
SC	-0.57	(0.14)	0.00
SD	-0.58	(0.14)	0.00
TN	-0.56	(0.14)	0.00
TX	-0.58	(0.24)	0.03
UT	-0.57	(0.14)	0.00
VA	-0.56	(0.14)	0.00
VT	-0.57	(0.14)	0.00
WA	-0.60	(0.21)	0.02
WI	-0.61	(0.14)	0.00

Notes: Data: NVSS birth records, SEER county population counts, NDCP childcare prices, hand-collected maximum group size data, for 2010-2019, 2021-2022. This table shows the estimated IV effects of log childcare prices on the log of the birthrate per 1000 women aged 20 to 49 for the full sample of states (“All states”), and for subsamples where I drop one state at a time. The childcare price is the median weekly price for full-time care at a childcare centre averaged for 0 to 2 year olds, winsorised at 99th percentile. The instrumental variable is the state-level maximum group size average for 0 to 2 year olds. All models control for county median earnings, the unemployment rate, female labour force participation, the male-female ratio, a housing price index, racial and ethnic composition, and county and year fixed effects. Estimates weighted by births to women aged 20 to 44. Standard errors are clustered at the state-year level.

Appendix Table D12. Robustness of IV estimates to winsorising

	20-44	20-24	25-29	30-34	35-39	40-44
<i>95 percentile</i>						
Log (Childcare Price)	-0.47*** (0.15)	-0.12 (0.16)	-0.10 (0.17)	-0.59*** (0.17)	-0.69*** (0.20)	-0.62*** (0.15)
<i>98 percentile</i>						
Log (Childcare Price)	-0.54*** (0.15)	-0.14 (0.18)	-0.12 (0.19)	-0.70*** (0.16)	-0.83*** (0.21)	-0.76*** (0.18)
<i>99 percentile</i>						
Log (Childcare Price)	-0.57*** (0.14)	-0.14 (0.18)	-0.12 (0.20)	-0.74*** (0.15)	-0.88*** (0.21)	-0.81*** (0.19)
<i>No winsorising</i>						
Log (Childcare Price)	-0.69*** (0.18)	-0.16 (0.20)	-0.15 (0.23)	-0.91*** (0.19)	-1.11*** (0.27)	-1.03*** (0.28)
Mean	69.76	97.57	122.42	89.99	37.71	7.45

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: $N = 7,495$. Data: NVSS birth records, SEER county population counts, NDCP childcare prices, hand-collected maximum group size data, for 2010-2019, 2021-2022. This table shows the estimated effects of log childcare prices on the log of birthrates per 1000 women for different levels of winsorising of prices. “20-44” is the log of the county-level birthrate per 1000 women aged 20 to 44, “20-24” the birthrate per 1000 women aged 20-24, and so on. “Childcare price” is the median weekly price for full-time care at a childcare centre averaged for 0 to 2 year olds, winsorised at the 95th percentile, 98th percentile, and 99th percentile and with no winsorising. All models control for county median earnings, the unemployment rate, female labour force participation, the male-female ratio, a housing price index, racial and ethnic composition, and county and year fixed effects. Estimates weighted by births to that age band. Standard errors, clustered at the state-year level, in parentheses.

Appendix Table D13. Robustness of IV estimates to instrument missing values

	20-44	20-24	25-29	30-34	35-39	40-44
<i>NA = 25</i>						
Log (Childcare Price)	-0.57*** (0.15)	-0.13 (0.19)	-0.12 (0.20)	-0.73*** (0.15)	-0.88*** (0.21)	-0.81*** (0.20)
<i>NA = 30</i>						
Log (Childcare Price)	-0.57*** (0.14)	-0.14 (0.18)	-0.12 (0.20)	-0.73*** (0.15)	-0.87*** (0.21)	-0.80*** (0.19)
<i>NA = 35</i>						
Log (Childcare Price)	-0.57*** (0.14)	-0.15 (0.17)	-0.13 (0.20)	-0.73*** (0.15)	-0.87*** (0.21)	-0.80*** (0.19)
Mean	69.76	97.57	122.42	89.99	37.71	7.45

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: $N = 7,495$. Data: NVSS birth records, SEER county population counts, NDCP childcare prices, hand-collected maximum group size data, for 2010-2019, 2021-2022. This table shows the estimated instrumental variable effects of log childcare prices on the log of birthrates per 1000 women for three different approaches to treating states with no maximum group size restrictions. “20-44” is the log of the county-level birthrate per 1000 women aged 20 to 44, “20-24” the birthrate per 1000 women aged 20-24, and so on. “Childcare price” is the median weekly price for full-time care at a childcare centre averaged for 0 to 2 year olds, winsorised at the 99th percentile. “NA=25” sets the maximum group size to 25 for states with no regulation, “NA=30” sets the maximum group size to 30 (baseline specification), and “NA=35” sets the maximum group size to 35. All models control for county median earnings, the unemployment rate, female labour force participation, the male-female ratio, a housing price index, racial and ethnic composition, and county and year fixed effects. Estimates weighted by births to that age band. Standard errors, clustered at the state-year level, in parentheses.

Appendix Table D14. Robustness of estimates to removing bad controls

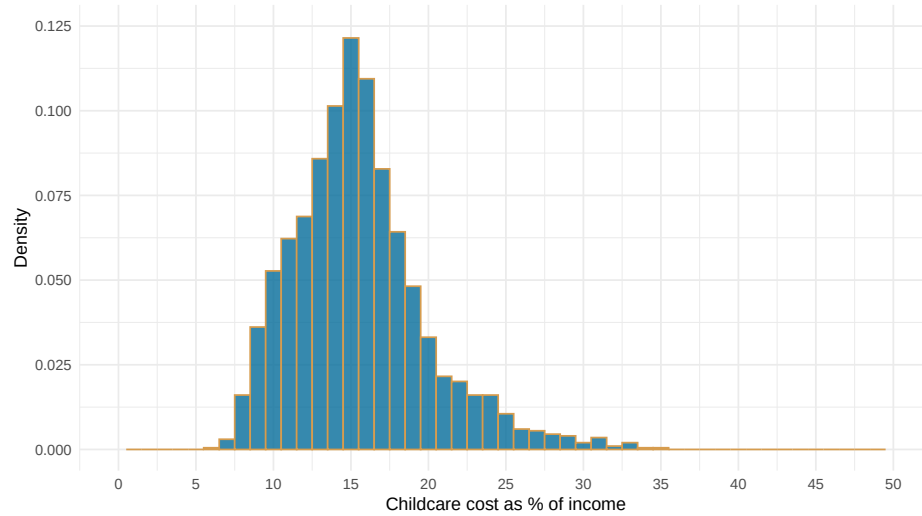
	20-44	20-24	25-29	30-34	35-39	40-44
<i>OLS</i>						
Log(Childcare Price)	0.01 (0.02)	0.02 (0.04)	0.03 (0.03)	0.01 (0.02)	0.01 (0.02)	-0.02 (0.03)
<i>IV</i>						
Log(Childcare Price)	-0.51*** (0.12)	-0.11 (0.16)	-0.04 (0.18)	-0.59*** (0.14)	-0.74*** (0.18)	-0.75*** (0.17)
<i>Reduced Form</i>						
Maximum Group Size	0.33*** (0.09)	0.07 (0.11)	0.02 (0.12)	0.38*** (0.09)	0.47*** (0.10)	0.47*** (0.08)
Mean	69.76	97.57	122.42	89.99	37.71	7.45

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: $N = 7,495$. Data: NVSS birth records, SEER county population counts, NDCP childcare prices, hand-collected maximum group size data, for 2010-2019, 2021-2022. This table shows the estimated effects of log childcare prices on the log of birthrates per 1000 women. Reduced form coefficients and standard errors multiplied by 100. “20-44” is the log of the county-level birthrate per 1000 women aged 20 to 44, “20-24” the log of the birthrate per 1000 women aged 20-24, and so on. “Childcare price” is the median weekly price for full-time care at a childcare centre averaged for 0 to 2 year olds, winsorised at the 99th percentile. “Maximum Group Size” is the state-level maximum group size average for 0 to 2 year olds. All models control for male county median earnings, the male unemployment rate, the male-female ratio, a housing price index, racial and ethnic composition, and county and year fixed effects. Estimates weighted by births to that age band. Standard errors, clustered at the state-year level, in parentheses.

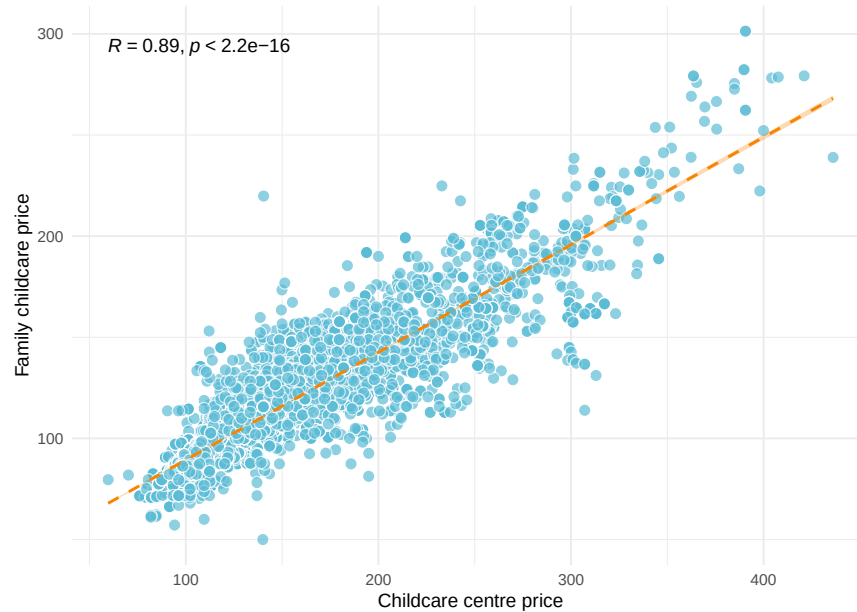
E Appendix Figures

Appendix Figure E1. Distribution of avg. annual cost of full-time centre-based childcare for 0-2 years as a share of median household income, 2010



Notes: Data: NDCP (including imputed data), American Community Survey, for 2010. The plot shows the distribution of the average annual cost for full-time care at a childcare centre for 0 to 2-year-olds, as a percentage of median household income at the U.S. county level.

Appendix Figure E2. Scatterplot of centre-based childcare and family childcare prices



Notes: N=11,789. Data: NDCP. This figure shows a scatterplot of the median weekly price for full-time care averaged for 0 to 2-year-olds, adjusted for inflation using base year 2010, at childcare centres and at family childcare homes. R is the Pearson correlation coefficient and p is the two-sided p-value from testing the null hypothesis that $R = 0$.



Day care concerns: Staffing standards in Nevada below national averages

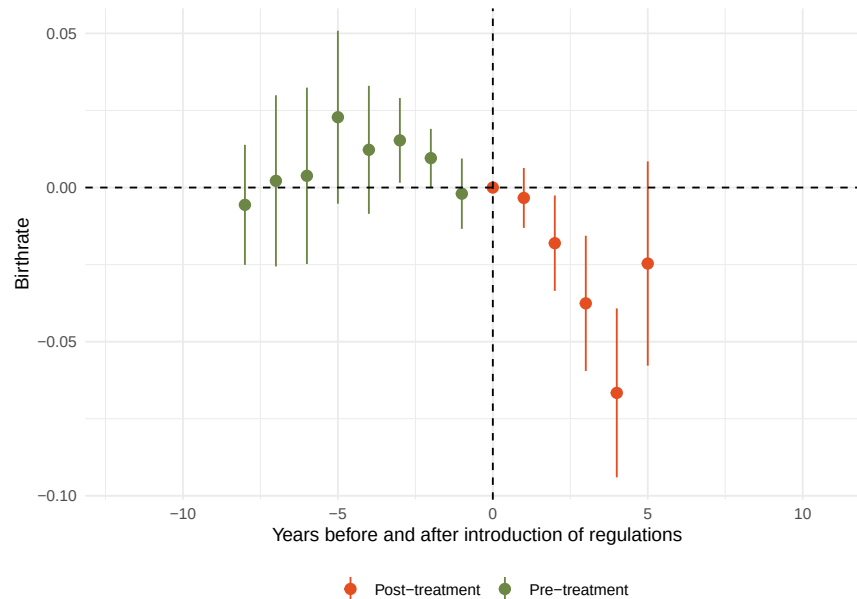
Thursday, May 15, 2003 | 11:24 a.m.

Nevada doesn't do as much as most other states to ensure that children are safe at day care centers, studies and experts say.

When it comes to child care staffing ratios, Nevada has some of the loosest regulations in the nation.

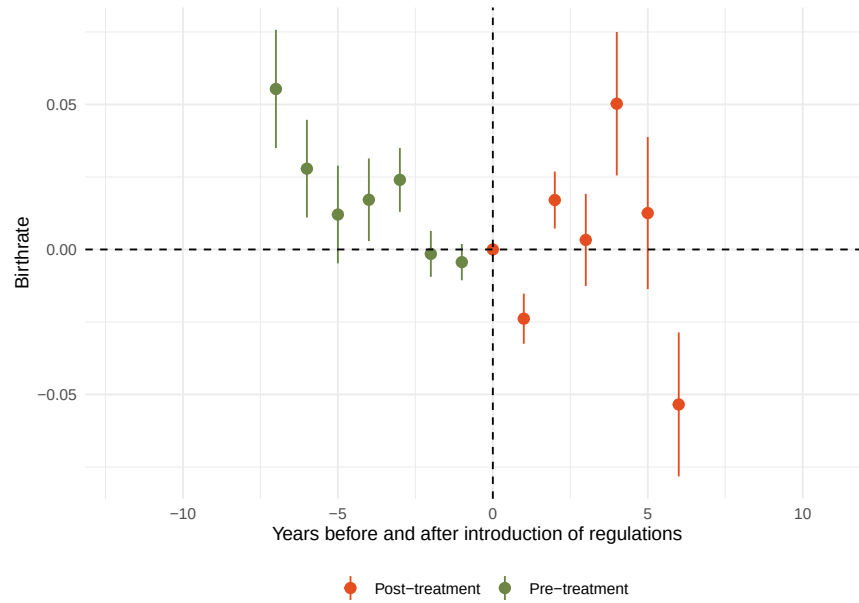
Nevada also ranks low in the number of licensed child care centers and in children served by government-subsidized child care, according to national comparisons. A state study last summer found that there were at least 6,000 children on waiting lists for child care.

Appendix Figure E4. Event study plot of childcare facility regulations on birthrates, Nevada



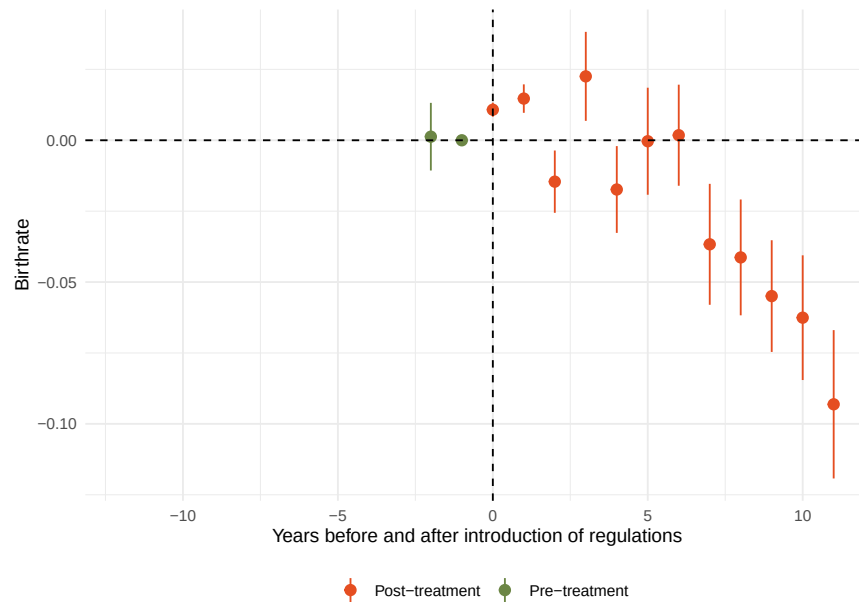
Notes: N=14,805. This figure shows an event study plot of effects of the introduction of childcare centre regulations in Nevada relative to control states on birthrates. The dotted vertical line is the year of treatment. The x-axis is relative time to the introduction of the regulations. The y-axis is the log birthrates for women aged 30 to 34. In 2017 NV introduced group size requirements for the first time and increased staff-child ratios for 18 month and 3-year-olds. The reference year is $t = 0$ as the regulations were introduced in September.

Appendix Figure E5. Event study plot of childcare facility regulations on birthrates, Vermont



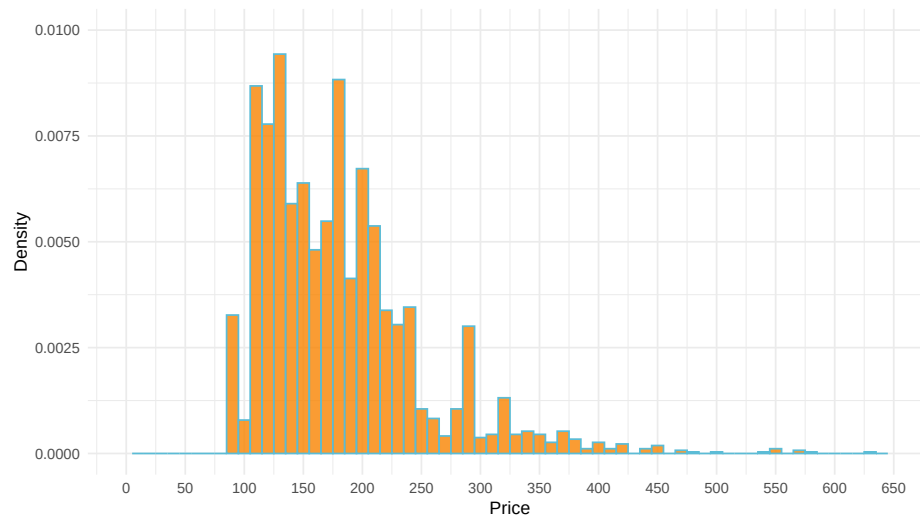
Notes: N=14,794. This figure shows an event study plot of effects of the introduction of childcare centre regulations in Vermont relative to control states on birthrates. The dotted vertical line is the year of treatment. The x-axis is relative time to the introduction of the regulations. The y-axis is the log birthrates for women aged 30 to 34. In 2016 VT relaxed group size requirements for 18 month olds. The reference year is $t = 0$ as the regulations were introduced in September.

Appendix Figure E6. Event study plot of childcare facility regulations on birthrates, Delaware



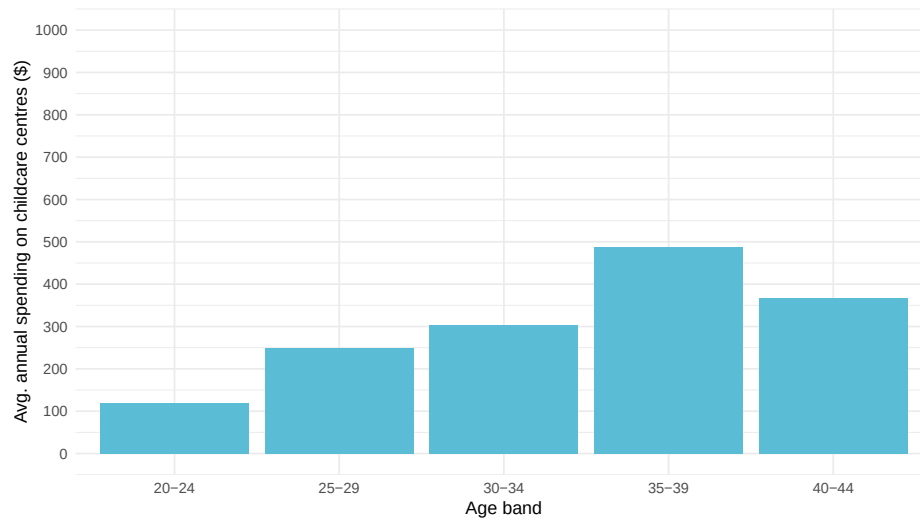
Notes: N=14,668. This figure shows an event study plot of effects of the introduction of childcare centre regulations in Delaware relative to control states on birthrates. The dotted vertical line is the year of treatment. The x-axis is relative time to the introduction of the regulations. The y-axis is the log birthrates for women aged 30 to 34. In 2011 DE introduced group size requirements for the first time. The reference year is $t = -1$ as the regulations were introduced in January.

Appendix Figure E7. Distribution of median price of full-time centre-based childcare for 0-2 years, 2022



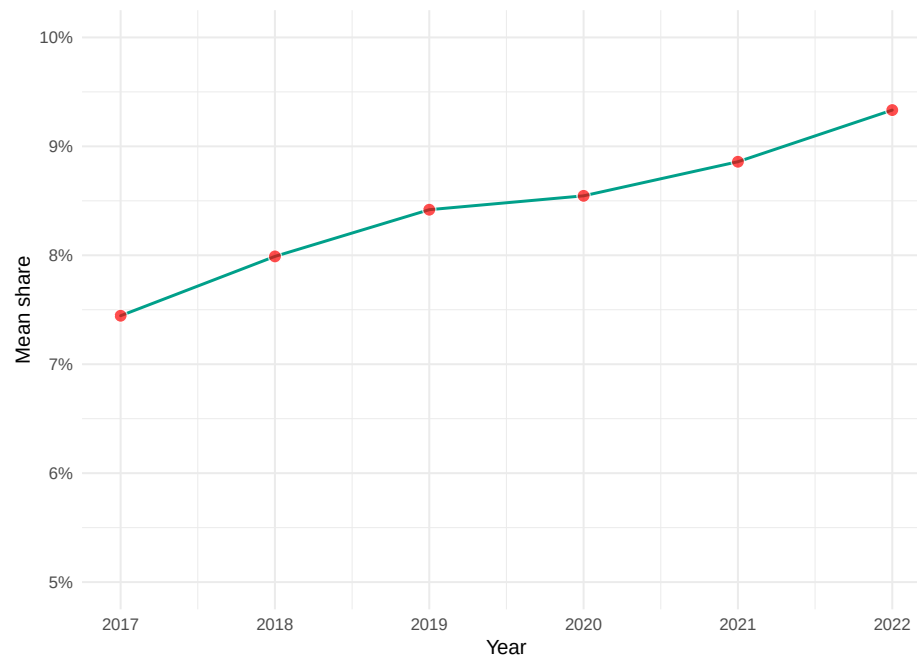
Notes: Data: NDCP (including imputed data), for 2022. The plot shows the distribution of the median price for full-time care at a childcare centre for 0 to 2-year-olds at the U.S. county level.

Appendix Figure E8. Average annual spending on childcare centres for 2010, by age



Note: Data: Consumer Expenditure Survey, for 2010. This figure shows the average annual spending on childcare centres by age of the respondent, for respondents with any children under 3. Childcare centres defined as day care centres, nurseries, and preschools.

Appendix Figure E9. Mean share of NAEYC accredited programmes over time



Notes: N=30. Data: NAEYC. This figure shows the mean share of NAEYC accredited programmes at the state level between 2017 and 2022.

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